Scaling Education Reform: Evidence from Morocco's Pioneer School Program*

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Can governments sustain large learning gains when evidence-based interventions move from pilot implementation to national scale? We study this question using Morocco's Pioneer School Program (PSP), a nationwide reform that combines two Global Education Evidence Advisory Panel "great buy" approaches: structured pedagogy and targeted remediation by learning level. The reform was rolled out in phases, expanding from 626 schools in 2023-24 to 4,626 schools in 2025-26, which allows us to compare effects across exposure duration and rollout cohorts. We estimate intent-to-treat effects using a matched-pair difference-in-differences design built from rich administrative records on all public primary schools and original student assessment data in Arabic, French, and mathematics. Student achievement is measured using psychometrically linked test scales (IRT), and identification is strengthened with post-double-selection LASSO controls and pre-trend evidence from seven years of historical data.

Three findings emerge. First, PSP generates large gains in foundational learning relative to comparison schools. Second, treatment effects vary across cohorts and years of exposure, consistent with changes in implementation conditions during rapid scale-up. Third, estimated impacts remain large in global comparative perspective, placing Morocco's reform among the higher-effect government-led interventions in recent evidence syntheses. Motivated by these patterns, we use the voltage-effect framework (List, 2026) to interpret how implementation quality, sample composition, and effective dosage can mediate the pilot-to-scale transition. The results contribute to the economics of education reform by showing both the promise of system-level scaling and the constraints that determine whether early gains persist as reforms expand through public delivery systems.

Keywords: Educational Reform; Scaling; Human Capital; Morocco.

JEL classification: I20, I21, I28, O15, O22.

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1 Introduction

Low learning levels remain one of the central development challenges in low- and middle-income countries. More than half of children are unable to read with comprehension by age 10, and deficits in foundational numeracy are similarly widespread (World Bank, 2019; Angrist et al., 2021). The policy challenge is therefore not only to identify interventions that work in small pilots, but to implement reforms through public systems at a scale large enough to change aggregate human capital trajectories (World Bank, 2018; Allcott, 2015; Vivalt, 2020).

This paper studies that challenge in Morocco, where the government launched the Pioneer School Program (PSP) as the core basic-education reform under the 2022–2026 roadmap. The PSP combines two intervention families emphasized by recent global evidence: structured pedagogy and targeted remediation by learning level (GEEAP, 2023; Banerjee et al., 2017; Duflo et al., 2024; Kerwin and Thornton, 2021; Piper and Dubeck, 2024). Unlike stand-alone NGO programs, PSP is a system-level reform embedded in ministry delivery structures, including teacher specialization, inspector-led pedagogical support, school-level certification, and digitalized assessment routines.[1]

When evaluating any program, credible estimates of the program's impact require a valid counterfactual (viz., comparison group). Our study relies on a difference-in-differences strategy. Using administrative data on all public primary schools in Morocco, Ibrahim et al. (2025) applied machine learning methods to select a similar non-PSP primary school for each of the 626 PSP schools that implemented the PSP in September of 2023. From these 626 pairs of schools, 138 of the closest matched pairs were selected for the study of Ibrahim et al. (2025). We call these the Cohort 1 schools. In this study we follow 92 pairs of these Cohort 1 schools for a second year (46 pairs were dropped because the comparison school in those pairs implemented the PSP program in September of 2024). We then supplemented these 92 pairs with 31 pairs of schools which include one school that started the PSP in September of 2024 and a similar comparison school that was matched to the treatment school using the same machine learning methods that were used by Ibrahim et al. (2025). We call these schools the Cohort 2 schools. Furhter, we follow another 75 pairs with an even split between PSP starting in September of 2025 and comparision schools using the same methodology. We call these schools Cohort 3. We collected information for grades 1 through 6 – primary school levels. To estimate the PSP program's impact in its second year, we collected primary assessment data and compared changes in student learning over time in the PSP schools to the same changes for the matched non-PSP schools, separately for the two cohorts of schools. For both cohorts, the key assumption is that the average (conditional) trend in the PSP schools would have been the same as the average (conditional) trend in the matched non-PSP schools had the PSP schools not participated in the program. For Cohort 1 schools, comparisons of trends in exam scores over the seven years before the program was implemented support this "parallel trends" assumption. For the Cohort 2 schools, similar comparisons for the same seven years before the start of the PSP program also support this assumption.

[1]The criteria for school recognition in the PSP is a ministerial commission label awarded as 'the École Pionnière' (Pioneer School). Certification is based on the ONDH (National Observatory of Human Development) evaluation reports covering three pillars: (1) Students: TaRL implementation and the remediation period, (2) Teachers: Specialization, training, certification, and one full year of implementation monitored by inspectors, (3) Schools: Infrastructure and broader program-level requirements.

The scale-up was rapid. The program began in 626 public primary schools in September 2023, expanded to 2,626 schools in September 2024, and again to 4,626 schools in September 2025. This rollout creates variation in both timing and exposure duration that is directly informative about an economic question at the center of the scaling literature: do gains persist, attenuate, or grow as implementation moves from early rollout conditions to broader administrative coverage?

Our empirical design uses two complementary data sources. First, we use national administrative records covering all public primary schools in Morocco over multiple years. Second, we use original baseline and endline student assessments in Arabic, French, and mathematics from matched PSP and non-PSP schools. We estimate student learning with linked item-response-theory (IRT) scales so that performance is comparable across grades, subjects, and survey rounds.

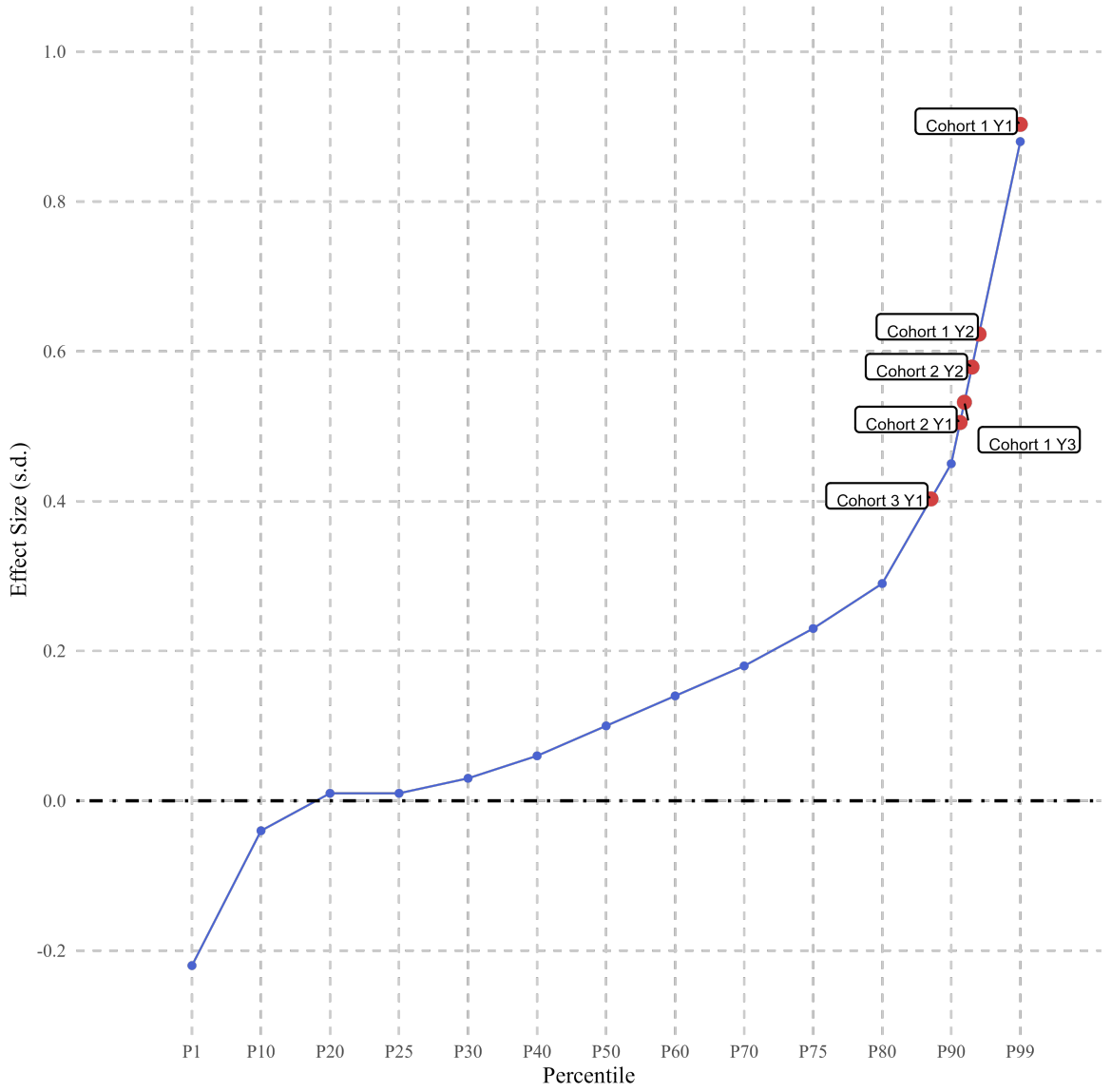
Because treatment assignment is not random, identification relies on matched-pair difference-in-differences. Schools are matched using high-dimensional predictors selected with post-double-selection LASSO, and treatment effects are identified from differential changes in outcomes between PSP schools and their matched comparison schools. We assess the credibility of the design using pre-program trend evidence from historical achievement data and report inference with clustering at the matched-pair level.

The main results show large and statistically precise gains in student learning from PSP participation. Effect sizes are substantial across Arabic, French, and mathematics, and large relative to the global distribution of education intervention effects (Figure 1). At the same time, the trajectory of effects differs across cohorts and years of exposure, consistent with the idea that implementation conditions changed during scale-up. These dynamics motivate a structured interpretation of scaling constraints rather than a binary "works/does not work" conclusion.

To organize that interpretation, we use the voltage-effect framework (List, 2026), which decomposes pilot-to-scale differences into mechanisms such as implementation quality, sample composition, and effective dosage. Our setting is well suited to this exercise because we observe the same reform as it transitions from an initial rollout to broader expansion under government delivery.

This paper contributes in three ways. First, it provides evidence on a large government-led reform that bundles pedagogical and organizational changes, speaking to policy design for foundational learning and human capital accumulation. Second, it contributes to the econometrics of real-world rollouts by combining machine-learning-assisted matching with difference-in-differences in a setting with evolving sample composition. Third, it links reduced-form impact estimates to a structural scaling framework, offering an interpretable bridge between short-run effect sizes and the broader question of whether education reforms can maintain performance at national scale.

Figure 1: *Comparison of ITT Effects to Global Evidence of Impacts from Education Interventions*



Note: This figure shows the distribution of effects reported in Evans and Yuan (2022). The estimated ITT effects for each cohort and year of exposure to PSP are overlayed onto this figure. Estimates come from Table 2.

2 Context

2.1 Learning Crisis in Morocco

More than half of children in low- and middle-income countries do not learn to read with comprehension by age 10 (World Bank, 2019). Similarly, poor performance has been found in students' basic mathematics skills (Angrist et al., 2021). The World Bank (2018) describes this state of affairs as a global "learning crisis". While nearly all of these countries have enrolled almost all of their children in primary school, progress in terms of student learning in these schools is often disappointing. What can be done to increase student learning in these countries? Recent research has provided evidence on which educational interventions increase student learning in low- and middle-income countries. The Global Education Evidence Advisory Panel (GEEAP, 2023) conducted a systematic review of the evidence to identify "great buy" interventions, which are defined as interventions that have been found to be highly cost-effective based on a large body of evidence. Unfortunately, many promising interventions have been found to be less effective when program implementation is transferred from non-governmental organizations to governments (Vivalt, 2020; Allcott, 2015). This observation holds for two of the above-mentioned "great buy" recommendations: scripted lesson plans (Kerwin and Thornton, 2021; Piper and Dubeck, 2024) and targeting instruction to a child's learning level (Banerjee et al., 2017; Duflo et al., 2024).

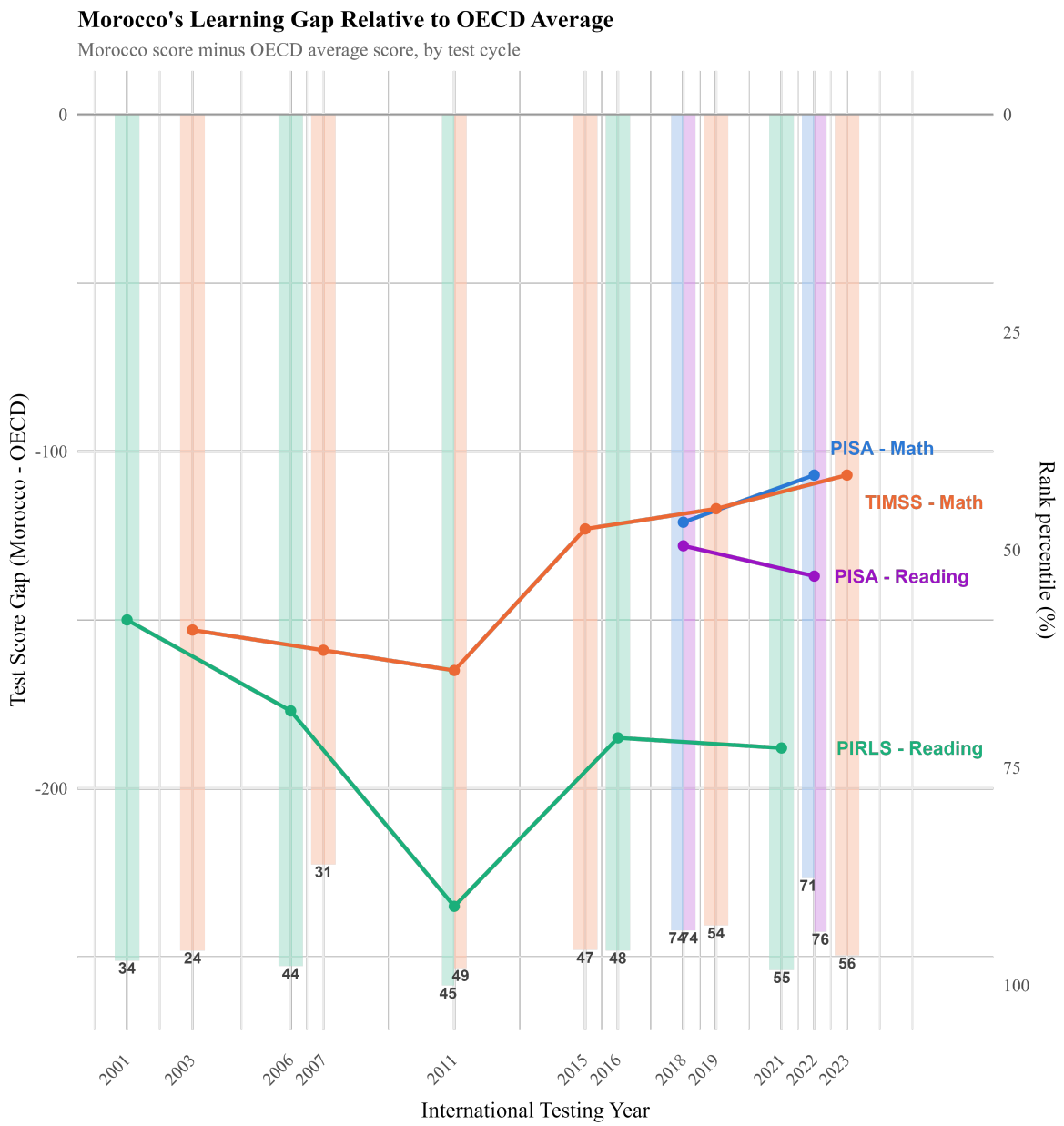
Public provision of education, as governed by the Ministry of Education (MENPS or Ministère de l'Education Nationale, du Préscolaire et des Sports), is the most common type of primary school education in Morocco. While the share of private enrollment has increased in recent years, as of 2022, public schools still enrolled 83 percent of primary school students in Morocco. Primary school enrollment numbers are high; the net enrollment rate is 99 percent. Persistence is also high; as of 2021, 98 percent of enrolled primary school students persisted to grade 6, the last grade of primary school (UIS-UNESCO, 2024).

However, two major concerns for education in Morocco are low continuation rates into secondary school and low student performance in both primary and secondary school. Only about 70 percent of Moroccan students who start primary school transition from primary to secondary education (Mansouri and Moumine, 2017). Equally – if not more – concerning are low levels of student learning. In the 2021 PIRLS reading assessment, despite recent progress, Morocco's fourth-graders ranked second to last (out of 57 countries), and more than half of the students (59 percent) did not even reach the minimum proficiency benchmark (PIRLS, 2021). Similarly, in the 2019 TIMSS mathematics assessment of fourth grade students, Morocco ranked among the lowest-performing countries (60 out of 64), and more than half of the students (57 percent) did not even reach the minimum proficiency benchmark (Mullis et al., 2019).

Morocco has particularly struggled to address the learning crisis for primary education. Figure 2 displays the trend in Moroccan international test performance over the past two decades. Compared to the OECD (rich country) test average performance, Morocco consistently is scoring roughly 100 to 200 points worse. TIMSS, PISA, and PIRLS all structure their exams as having roughly a 500 point average with 100 point standard deviations. This means that Morocco is consistently one to two standard deviations away from the mean primary student performance – well below the international average. Rank ordered relative to the national

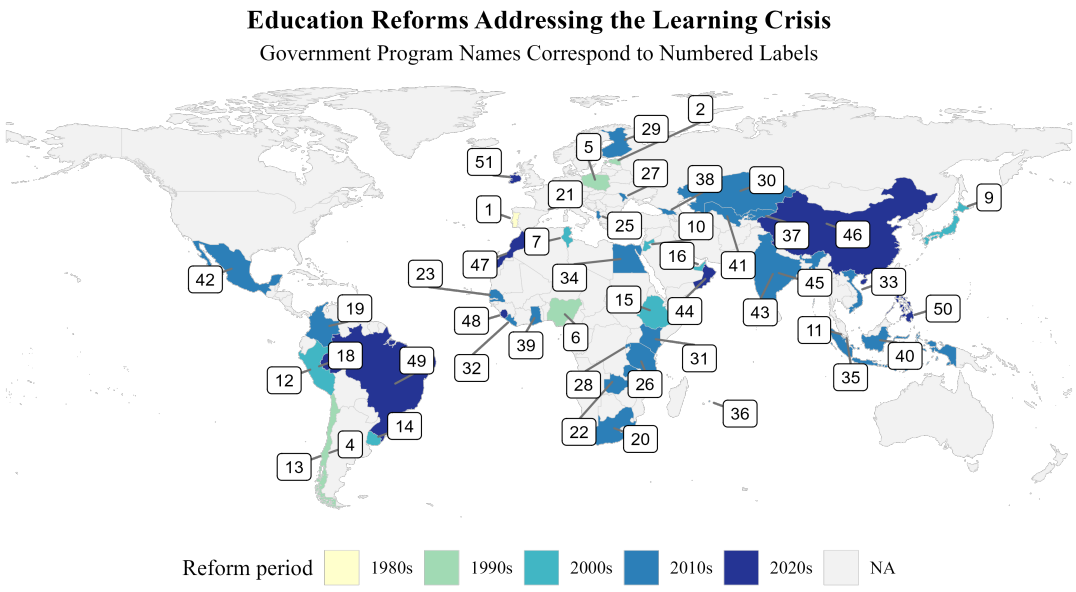
performance of other countries participating in these exams, Morocco has ranked near last in almost every exam they have participated in. The primary student performance on these exams places them in a percentile which is never below the 80th percentile. In other words, 80 percent of primary school students outperform the average Moroccan student and in the majority of years Moroccan students perform worse than 95 percent of all primary school students.

Figure 2: *Morocco's International Test Performance*



Note: This figure demonstrates the low performance of Morocco on international student tests. The line graphs represent the difference between Morocco's average test score and the OECD average test score. The bars depict the percentile Morocco's county rank places amongst all national averages. The labels on the bars provide the country rank of Morocco across all country averages. The TIMSS math scores and PIRLS reading scores are tested on 4th graders. PISA math and reading scores are tested on 15 year olds.

Figure 3: *Government Education Reforms Addressing the 'Learning Crisis'*



Note: This map displays education reforms enacted by governments. The coloring associates to the decade of the reform. And the numbered labels correspond to the numbered reforms in Table **??**.

Table 1: *Education Reforms Corresponding to Map Labels*

Label	Country	Education Reform	Year	Label	Country	Education Reform	Year
1	Portugal	Lei de Bases do Sistema Educativo	1986	29	Finland	National Core Curriculum	2016
2	Estonia	National Framework Curriculum	1996	30	Kazakhstan	Updated Curriculum	2016
4	Chile	Jornada Escolar Completa	1997	31	Kenya	CBC 2-6-6-3	2017
5	Poland	6+3+3	1998	32	Liberia	Getting to Best Education Sector Plan	2017
6	Nigeria	Universal Basic Education	1999	33	Vietnam	New General Education Curriculum	2018
7	Tunisia	Loi d'orientation de l'éducation	2002	34	Egypt	Education 2.0	2018
9	Japan	Yutori Kyoiku	2002	35	Singapore	Learn for Life	2018
10	Jordan	ERfKE	2003	36	Mauritius	Nine Year Continuous Basic Education (NYCBE) framework	2018
11	Singapore	Teach Less, Learn More	2004	37	Kyrgyzstan	Altyn Kazyk	2018
12	Peru	Carrera Pública Magisterial (CPM)	2007	38	Georgia	New School Model	2018
13	Chile	Preferential School Subsidy (SEP)	2008	39	Ghana	Standards-Based Curriculum	2019
14	Uruguay	Plan Ceibal	2008	40	Indonesia	Merdeka Belajar	2019
15	Ethiopia	General Education Quality Improvement Programme (GEQIP)	2008	41	Uzbekistan	National Education Development Concept	2019
16	United Arab Emirates	Madares Al Ghad	2010	42	Mexico	Nueva Escuela Mexicana (NEM)	2019
18	Peru	Ley de Reforma Magisterial (LRM)	2012	43	India	National Education Policy (NEP)	2020
19	Colombia	Todos a Aprender	2012	44	Oman	Education Vision 2040	2021
20	South Africa	Curriculum and Assessment Policy Statement (CAPS)	2012	45	India	NIPUN Bharat	2021
21	Andorra	Educational System Reform (PERMSEA)	2013	46	China	Double Reduction Policy	2021
22	Zambia	Zambia Education Curriculum Framework	2013	**47**	**Morocco**	**Écoles Pionnières**	**2022**
23	Senegal	PAQEEB	2013	48	Sierra Leone	Basic and Senior Secondary Education Act	2023
25	Albania	Pre-University Education Strategy	2014	49	Brazil	National Education Plan (PNE)	2025
26	Tanzania	Education and Training Policy	2014	50	Philippines	EDCOM 2	2026
27	Moldova	Education Code	2014	51	Ireland	Delivering Equality of Opportunity in Schools (DEIS)	2026
28	Rwanda	CBC	2015				

2.2 The Pioneer School Program (PSP): A Targeted Education Reform

The Pioneer School Program (Programme des Écoles Pionnières, PSP) is a systemic reform initiative implemented by the Ministry of National Education, Preschool and Sports (MENPS) as the operational vehicle for the 2022–2026 Roadmap for a Quality Public School.

The program's statutory objectives are threefold:

1. **Foundational Learning:** Improvement of student mastery in Arabic, French, and Mathematics.
2. **Dropout Reduction:** Reduction of school dropout rates (targeted reduction of 30%).
3. **Student Fulfillment:** Enhancement of soft skills and socialization through extracurricular activities.

From the perspective of this evaluation, the PSP is treated as a multi-component intervention acting simultaneously on three axes:

- **The Student:** Intensive remediation via Teaching at the Right Level (TaRL) and classroom restructuring via Explicit Teaching.
- **The Teacher:** Competency-building through specialized training, subject specialization, and provision of scripted lessons and digital resources.
- **The School:** Voluntary adhesion to the program, upgrading of infrastructure and equipment, and implementation of a performance-based Projet d'Établissement Intégré (PEI).

2.2.1 Targeted Remediation (TaRL)

Students undergo diagnostic assessments (ASER/TaRL tests). Data is entered into Massar, which uses an algorithm to sort students into proficiency groups (Paliers) rather than age or grade.

1. Example Levels (Arabic/French): Beginner, Letter, Word, Paragraph, Story, Comprehension.
2. Example Levels (Math): Number recognition (0-9, 10-99), Operations (Addition, Subtraction, etc.).

Implementation includes:

1. A 4-to-6-week period (depending on the cohort) at the start of the year where the regular syllabus is suspended to focus exclusively on foundational skills,
2. Continued remediation sessions (Soutien scolaire) held outside standard hours for students who have not yet reached grade level.

2.2.2 Structured Pedagogy

This component replaces traditional instruction in Arabic, French, and Mathematics.

Implementation follows a structured sequence:

1. Presentation of objectives,
2. Modeling ("I do"),
3. Guided practice ("We do"),
4. Independent practice ("You do"),
5. Closure and objectivization.

Instruction relies on standardized scripted lessons (fiches pédagogiques) and digital supports (PPT). Approximately 300,000 standardized slides were produced for classroom projection. Instructional materials are aligned with the 2022 curriculum revision, which streamlined content to prioritize foundational skills (e.g., introducing fractions earlier in Math).

2.2.3 Teacher Specialization

Where staffing allows, schools organize teaching into subject poles:

- Pole 1: Arabic, Islamic Education, History-Geography,
- Pole 2: French,
- Pole 3: Mathematics and Scientific Activity.

2.2.4 School Recognition

Schools are audited against a reference framework to obtain the "École Pionnière" Label (Decree n° 2.24.144).

Certification triggers:

- Individual: Teaching staff certified in PSP practices receive a gross annual performance bonus of 10,000 MAD. Note: The structure of this incentive (bonus vs. salary integration) was subject to negotiation following the 2023-2024 strikes,
- Collective: Labeled schools receive an additional operating budget (up to 100,000 MAD) via the PEI to improve school infrastructure and hygiene.

2.2.5 School Management and Digital Ecosystem

Schools must execute a Projet d'Établissement Intégré (PEI) approved by the Provincial Directorate, setting quantitative targets for learning and dropout reduction.

The reform includes large-scale deployment of digital equipment (laptops for teachers, overhead projectors and dedicated projection space in every classroom) and systematic entry of assessment data into Massar.

2.3 Role of Inspectors and Pedagogical Supervision

Within the PSP framework, inspectors transition from an administrative compliance role (grading for promotion) to a pedagogical coaching model.

Mandate Evolution The "Mentor" Model: Inspectors are designated as the primary "messengers" of the reform. They manage a restricted portfolio with a contractual target of weekly visits to support teachers in the classroom. Unlike the standard inspectorate, PSP inspectors receive specific training, digital equipment, and a monthly indemnity to compensate for the increased frequency of field visits and reporting requirements.

Operational Functions Inspectors are the operational pivot of the program, executing four key functions:

1. Cascade training of teachers: Acting as master trainers, 157 inspectors (Wave 1) were trained by international experts (Gauthier Bissonnette) and subsequently trained approx. 10,700 teachers,
2. Pedagogical Supervision: Conducting classroom observations using standardized grids to monitor fidelity to the Explicit Teaching model and TaRL routines,
3. Data Verification (Quality Assurance): Inspectors act as external auditors for learning data. They verify a random sample of student test copies and validate the scores entered by teachers into Massar to ensure the reliability of the national dashboard (calculating concordance rates between teacher and inspector grading),
4. Certification: They evaluate teacher performance against the PSP framework to validate the "Pioneer Teacher" certification, which triggers financial bonuses.

While the protocol dictates weekly support, Wave 1 monitoring data (INE, May 2024) reveals significant heterogeneity in execution:

1. Visit Frequency: The target of weekly visits is not uniformly met. While 90.7% of schools in the Oriental region reported weekly visits, this figure drops to 22.7% in Guelmim-Oued Noun and 36.8% in Drâa-Tafilalet.

2. Reporting Gaps: 47.6% of schools reported receiving zero written inspection reports in the three weeks prior to the survey.

3. Structural Constraints: Qualitative field reports indicate that in some non-PSP or transition contexts, inspector-to-teacher ratios remain critically high (up to 1:180), forcing inspectors to prioritize administrative emergencies over pedagogical coaching.

3 Data

This study uses two sources of data. The first is administrative data for all of Morocco's public primary schools. The second is baseline and endline assessment data that measure students' performance in mathematics, Arabic and French for 338 public primary schools.

3.1 School Sampling and the Rollout of PSP

The Pioneer School Program (PSP) has evolved substantially since its launch. Rather than a fixed intervention, it has undergone structural adjustments in scale, governance, instructional intensity, monitoring architecture, and incentive design across successive phases.

3.2 Phase 1 (2023–2024): Experimental Pilot – Cohort 1

The first year constituted a large-scale experimental phase implemented in 626 primary schools across all 12 regions.

This phase was characterized by:

- A high-intensity instructional model, including a 6-week intensive remediation period at the beginning of the academic year;
- A tightly conditional incentive structure linking individual teacher bonuses (10,000 MAD annually) to both school-level labeling and individual certification;
- Direct inspector-led cascade training and close pedagogical supervision;
- Initial logistical constraints, including delays in the delivery of textbooks and digital projection equipment in some regions.

Implementation during this phase coincided with nationwide teacher strikes lasting approximately three months, which disrupted instruction and created regional heterogeneity in exposure and implementation fidelity.

Phase 2 (2024–2025): Rapid Expansion – Cohort 2

The second phase marked a transition from experimental pilot to systemic scaling.

Primary-level coverage expanded from 626 to approximately 2,626 schools, representing roughly 30% of public primary enrollment. The reform simultaneously extended to the lower secondary cycle through the launch of 232 "Collèges Pionniers."

This expansion was accompanied by several structural adjustments:

- A formal reduction in the duration of the intensive remediation period (from 6 to 4 weeks for newly entering schools);
- Strengthening of logistical coordination, with improved availability of instructional materials at the start of the academic year;
- Reconfiguration of the incentive regime following the resolution of the 2023–2024 social unrest.

In particular, the originally PSP-specific conditional performance bonus was largely integrated into a broader nationwide salary adjustment for teachers. As a result, in Year 2 the marginal financial incentive specific to PSP participation became less distinct from the general compensation framework, potentially altering incentive salience and motivation dynamics.

The expansion phase also required scaling of supervisory capacity, with a substantial increase in the number of inspectors and teachers operating under the PSP framework.

Phase 3 (2025–2026 onward): Progressive Generalization

Beginning in 2025–2026, the PSP entered a progressive generalization phase, with the addition of approximately 2,000 primary schools per year.

The medium-term objective is to reach approximately 50% coverage of public primary establishments by 2025–2026, with full national generalization (approximately 8,630 establishments) projected for the 2027–2028 academic year.

Full national coverage of primary establishments (approximately 8,630 schools) is projected by 2027–2028. Due to the national expansion in September 2024 (Phase 2), a subset of Control schools from Cohort 1 were selected by the MENPS to become Pioneer Schools.

3.3 Student Performance: Item Response Theory (IRT)

Morocco's Ministry of National Education, Preschools and Sports granted the authors access to historical administrative data on all 8,250 public primary schools[2] in Morocco over a 7-year period, from 2015 to 2022. The administrative data include more than 248 variables

[2]While there are 8,250 central public primary schools in Morocco, we consider all satellite schools attached to the mother (central) schools for a universe of 21,077 schools.

measuring student performance and socioeconomic status for 24 million students, as well as many variables on the characteristics of approximately the 104,000 teachers in these schools, and on the characteristics of the 21,077 schools themselves.

This study's main outcomes of interest are student performance on academic assessments in Arabic, French, and mathematics. For the first year of data collection for Cohort 1, baseline assessments for each of these three subjects were administered in September of 2023, and endline assessments were administered in June and July of 2024. For the second year of data collection, baseline assessments for Cohort 2 were administered in September of 2024, and 1-year endline assessments for Cohort 2, along with 2-year endline assessments for Cohort 1, were administered in June and July of 2025. These assessments record students' responses for all test questions (test "items"). For each subject, a two-parameter logistic (2PL) item response theory (IRT) model (De Ayala, 2013; Kolen and Brennan, 2014) was estimated to aggregate these responses and generate continuous estimates of student performance, using overlapping items (or "anchors') to map test scores onto common scales (across grades as well as across the baseline and endline data collection rounds). In doing so, differential item functioning was investigated, and only test questions with stable item parameters were retained. Each subject's continuous test score was standardized by subtracting the mean, and then dividing by the standard deviation, of the distribution of the test scores at endline of the students in the matched (non-PSP) schools. Further details on the composition and quality of the assessments are elaborated upon in Appendix Section **??**.

The study's sample of schools was constructed in four steps, using administrative data on the universe of government schools. We performed these steps separately for the Cohort 1 and Cohort 2 samples. The first cohort (Cohort 1) are the original PSP schools and comparison schools selected for the year 1 assessment over the 2023 – 24 academic year. The second cohort (Cohort 2) are the new set of PSP schools and comparison schools selected for the year 2 assessment over the 2024 – 25 academic year. The main difference in these two cohorts of schools are the timing of the rollout of the PSP reforms. The PSP schools in Cohort 1 implemented the program in September of 2023, and so they had two years of exposure to the PSP in July of 2025, while the PSP schools in Cohort 2 implemented the program in September of 2024, and so they had only one year of exposure in July of 2025.

4 Methodology

4.1 Matched Paired PDS LASSO

To improve the viability of the conditional parallel trends assumption, we matched schools using the best predictors of treatment status and baseline outcome (Ham and Miratrix, 2024). With access to a large administrative dataset on all primary schools in Morocco, we were able to implement a machine learning algorithm to identify the most relevant predictors of treatment status and a baseline measure of the outcome of interest (Ham and Miratrix, 2024). The algorithm used is post-double selection LASSO (PDSLASSO) used for estimating structural parameters in linear models with many controls (Belloni et al., 2012, 2011, 2014, 2016) .

The PDSLASSO algorithm operates by performing two LASSO regressions. First, it estimates a LASSO regression with the baseline outcome variable as the dependent variable and a set of potential control variables as regressors (all 248 constructed variables in the database). Second, it performs another LASSO regression with the treatment indicator as the dependent variable and the same set of control variables as regressors. The final set of control variables included in the model is the union of the variables selected in these two steps. This approach ensures that the chosen controls are those most predictive of both the outcome and the treatment assignment, thereby enhancing the robustness of causal inference (Ahrens et al., 2019).

Using a school-level database with 248 variables, consisting of school-level characteristics and aggregated student and teacher characteristics in 8,034 primary schools (including the pre-selected program schools), the PDSLASSO implemented identified 16 variables to be used for matching.

Next, we implemented an iterative process of a calibrated Mahalanobis distance matching algorithm using the variables selected by the PDSLASSO, to match 626 untreated schools (from the universe of 7,408 untreated schools) to the 626 selected schools for treatment. After each iteration of matching with different distance calipers, we implemented post-matching diagnostics, including tests for common support, external validity, and balance between the treated and the matched untreated schools. External validity was investigated by comparing the probability density function (PDF) and the cumulative distribution (CDF) for the average baseline measure of the outcome of interest in matched, unmatched, and all schools. Balance in baseline characteristics between the matched treated and untreated schools was evaluated by checking for any statistically significant difference between the paired schools.

The Mahalanobis distance matching was implemented without replacement while restricting the algorithm to select one matched (nearest neighbor) untreated school from the full list of 7,408 untreated schools. Initially, without distance calibration, the matching algorithm successfully matched 539 treated schools (out of 626) with 539 untreated schools, resulting in a total of 1,078 schools. However, the matched untreated schools from this uncalibrated attempt had significantly different characteristics compared to their matched treated schools, based on the same variables selected by the PDSLASSO. To address this imbalance, we iteratively adjusted the Mahalanobis distance caliper in descending order, starting with the highest possible caliper. We evaluated the baseline balance between schools and examined the PDF and CDF of the baseline measure of the outcome of interest. This process continued until we identified a caliper that produced matched schools with balanced baseline characteristics.

The final caliper used in the matching exercise was root 11.5, producing a list of 496 matched pairs of schools (a total of 992 schools). The population of good matches from the previous step was then used to randomly sample the schools to participate in the baseline and endline data collection for the evaluation of the Pioneer School Program. The evaluation sample was agreed to include 150 treated schools and 150 untreated schools. The evaluation sample was randomly drawn, stratified by terciles of the final grade 6 exam score (the measure for the baseline outcome of interest) and the 12 regions in Morocco, resulting in a total of 36 strata. However, there were 6 strata belonging to the 3-baseline outcome terciles in 2 regions with very few schools. As a result, we only stratified the school sampling by region for those 2 regions, with a final total of 30 strata.

Of the 300 sampled schools, baseline data collection could not be conducted in 19 schools (one of which had permanently closed shortly before the study started). Five of the remaining 281 schools (three pioneer schools and two comparison schools) lost their matched schools due to this reduction of the sample to 281 schools. Dropping the five schools without a match resulted in the studys effective sample of 276 schools.

4.2 Difference-in-Differences

This section reports preliminary results of the impact of the PSP after three years. More specifically, it presents estimates of the intent-to-treat effect of the PSP intervention on student learning using a strategy that combines difference-in-differences estimation with matching methods. This strategy compares changes over time in the outcome variables of interest for primary schools that were assigned to receive that intervention with the same changes for their matched comparison schools. For all outcomes, the following empirical specification is used:

$$Y_{igsjt} = \beta(TREAT_{sj} \times POST_t) + \delta(X_{igsj} \times POST_t) + \zeta(\eta_j \times \theta_g \times POST_t) + \gamma_{igsj} + \varepsilon_{igsjt} \quad (1)$$

Here, Y_{igsjt} is outcome Y for student i in grade g in school s in matched pair j in period t. $TREAT_{sj}$ is the assignment of school s in matched pair j to receive the education intervention, and $POST_t$ is a dummy variable equal to 1 if the outcome Y is measured at endline (i.e., after the PSP was rolled out) and 0 if measured at baseline (i.e., before the PSP was rolled out). This is the canonical difference-in-differences design.

The $\gamma_{i,g,s,j}$ term is a student fixed effect that measures the impacts on Y_{igsjt} of all student characteristics, observed or unobserved, that do not change over time. However, the impacts of some student characteristics may change over time, so a vector of observed student characteristics, denoted by X_{igsj}, is added and multiplied by $POST_t$. The corresponding vector of coefficients for this interaction term, denoted by δ, measures the change in the impacts of these observed student characteristics over time. In other words, δ measures the contributions of observed student characteristics to changes in the outcome variables over time. Since nearly all students stay in the same school between baseline and endline data collection, student fixed effects also include school fixed effects, and so the X_{igsj} variables can also include observed school variables. The student variables available in the dataset are students' gender and whether the student was eligible for a social safety net program (measured at baseline). The school characteristics are school-level average test scores, region, average class size, and the student-to-teacher ratio (all measured at baseline). A LASSO (Least Absolute Shrinkage and Selection Operator) procedure was used to determine which $X_{igsj} \times POST_t$ interactions to include in the specification.[3] The three-way interaction $\zeta(\eta_j \times \theta_g \times POST_t)$ represents grade-by-matched pair interactions with the post-period indicator. Intuitively, these fixed effects account for how much, on average, a PSP student's matched-pair comparison peers

[3]More specifically, we first calculated each student's change in test scores between the baseline and endline assessments. Then, we residualized this change in test scores, subtracting the comparison group's grade-by-matched pair mean. Lastly, with these residuals, we use a post-double-selection LASSO to select relevant predictors X_{igsj} (Belloni et al., 2011).

(who are enrolled in the same grade in the matched non-PSP schools) learned between baseline and endline; these effects also apply to corresponding PSP students, which in effect implements the parallel trends assumption.

The coefficient of interest is β; it measures the intent-to-treat (ITT) effect of assignment to the program. In estimating ITT effects, we allow for the (plausible) possibility that the program was not fully implemented in at least some of the PSP schools (none of the non-PSP schools received the program). Following de Chaisemartin and Ramirez-Cuellar (2024) on clustering in paired and small-strata experiments, we cluster standard errors at the matched-school-pair level.

For β in Equation (1) to estimate the causal impact of the PSP program on the outcomes of interest, the identifying assumption is that, conditional on the "comparison" variables in that equation, the average trend in the outcomes over time in each PSP school would have been the same as the average trend in that school's matched non-PSP school if the PSP had not been implemented in that PSP school. Figures 8, 9 and 10 provide evidence supporting this assumption for Cohorts 1, 2 and 3, respectively: in the seven (school) years before the PSP program started (from 2015-16 to 2021-22), for each cohort changes over time in the Grade 6 exit exam scores are almost identical for the PSP schools and their associated comparison schools.

5 Results

5.1 Impacts on Test Performance

Table 2 presents the main results of this report for all three cohorts. The first four columns present the results for Cohort 1, which has been implementing the PSP for three years (since September of 2023). Column (5) - (8) present the results for Cohort 2, which has been implementing the PSP for two years (since September of 2024), and columns (9) - (12) present the results for Cohort 3, which has been implementing the PSP for only one year (since September of 2025). Finally, columns (13) - (16) present pooled results that combine cohorts, which will be explained in more detail below. Recall that these estimates are not based on IRT modeling, which was used in the previous reports and will also be applied in the final report. Instead, student performance is based on the percentage of correct answers, normalized using the mean and standard deviation of the students in the comparison schools, calculated for each assessment (test), separately for each grade, cohort and year (and separately for baseline and endline tests within any given year). Even so, the results here for Cohort 1 after two years of exposure and for Cohort 2 after one year of exposure are very similar to the IRT results provided in previous years.

Turning to Cohort 1, after one year of exposure, and combining all three exams, the PSP increased students' academic skills by 0.90 standard deviations (henceforth s.d.). The subject-specific impacts for Cohort 1 after one year of exposure are 0.49 s.d. for Arabic, 1.24 s.d. for French, and 0.99 s.d. for mathematics. After two years of exposure, the impacts are smaller than the impacts after one year of exposure: 0.62 s.d. for all three subjects combined, 0.36 s.d. for Arabic, 0.95 s.d. for French, and 0.61 s.d. for mathematics. This

decline was also seen in the Year 2 final report, yet they are all highly statistically significant and are very high in absolute terms. Using the distribution of impacts over 96 studies in developing countries compiled by Evans and Yuan (2022), the 0.36 s.d. impact for Arabic is slightly above the 80th percentile for reading, the 0.95 s.d. impact for French is above the 99th percentile, and the 0.61 impact for mathematics is at about the 95th percentile. Finally, after three years of exposure, the impacts for Cohort 1 are slightly smaller than the impact after two years of exposure: 0.53 s.d. for all three subjects combined, 0.24 s.d. for Arabic, 0.94 s.d. for French, and 0.54 s.d. for mathematics. Note that the impact for Arabic is statistically significant only at the 10 percent level, which is due to both the smaller estimate and the larger standard error due to the much smaller sample size. All other estimated effects for Cohort 1 are highly statistically significant for all three years. The reasons for this decline are still under investigation and will be discussed in detail in the final report.

Next, turn to the estimated impacts for Cohort 2, which started the PSP program in September of 2024; the results are shown in columns (5) - (8) of Table 1. These results could differ from those of Cohort 1 because Cohort 2 schools are somewhat different from the Cohort 1; for example, as seen in Table 6 (second and fourth columns), the Cohort 2 schools have a slightly higher repetition rate, a slightly higher percentage of students who qualify for the Tayssir program, and are slightly more likely to be in rural areas. After one year of exposure, the estimated impacts are: 0.51 s.d. for all three subjects combined, 0.30 s.d. for Arabic, 0.76 s.d. for French, and 0.49 s.d. for mathematics. These smaller impacts, compared to the estimates after one year of exposure for Cohort (1), were reported previously in the Year 2 Final Report. The reasons for this lower impact after one year of exposure are still being investigated; two possibilities are that the schools that started the PSP in the second year were not as motivated as those that started in the first year, and that the expansion of the program to a much larger group of schools (2,000, versus 626 in the first year) led to some difficulties in implementing the program. Yet these are still very large effects, and all are highly statistically significant; in terms of the results compiled by Evans and Yuan (2022), the 0.30 s.d. impact for Arabic is slightly below the 75th percentile for reading, the 0.76 s.d. impact for French is at about the 95th percentile, and the 0.49 s.d. impact for mathematics is at about the 95th percentile.

Moving to the results for Cohort 2 after two years of exposure, the estimated impacts are mostly larger than those after one year of exposure. More specifically, the increases in student performance were 0.58 s.d. for all three subjects combined, 0.31 s.d. for Arabic, 0.73 s.d. for French, and 0.58 for mathematics. These results are all statistically significant despite the relatively small sample size. This modest increase in the combined estimate, from 0.51 s.d. to 0.58 s.d., reflects very a small increase in Arabic (from 0.30 s.d. to 0.31 s.d.) and a large increase in mathematics (from 0.49 s.d. to 0.58 s.d.), and a modest decline in French (from 0.76 s.d. to 0.73 s.d.). This further, albeit modest, increase after a second year of exposure is what one would expect. Indeed, these estimated impacts for Cohort 2 after two years of exposure are almost as large as the estimated impacts for Cohort 1 after two years of exposure.

Finally, turning to Cohort 3. Again, These results could differ from those of Cohorts 1 and 2 because Cohort 3 schools are different from those in Cohorts 1 and 2; for example, as seen in Table 6 (fourth and sixth columns), the Cohort 3 schools have a much higher percentage of students who qualify for the Tayssir program, and are much more likely to be in rural areas. The estimated impacts for Cohort 3 in Table 1 after one year of exposure to the program are, except for mathematics, somewhat smaller than the estimated impacts for Cohort 2 after one

year of exposure: 0.40 s.d. for all three subjects combined, 0.25 s.d. for Arabic, 0.47 s.d. for French, and 0.50 s.d. for mathematics. Even so, these estimated impacts are all statistically significant and are quite large; in terms of the results compiled by Evans and Yuan (2022), the 0.25 s.d. impact for Arabic is at the 70th percentile for reading, the 0.47 s.d. impact for French is slightly below the 90th percentile, and the 0.50 s.d. impact for mathematics is at about the 95th percentile.

The last four columns (13) - 16) present results that combine cohorts, which produce estimates that are averages over cohorts. Panel A presents the average impact of the PSP program after one year of exposure, averaged over all three cohorts. The estimated impacts are 0.60 s.d. for all three subjects combined, 0.35 s.d. for Arabic, 0.81 s.d. for French, and 0.66 s.d. for mathematics. All of these estimates are highly significant and are very large effects. Panel B presents the average impact of the PSP program after two years of exposure, averaged over Cohorts 1 and 2. The estimated impacts are very similar to the averages after one year of exposure for all three cohorts: 0.61 s.d. for all three subjects combined, 0.36 s.d. for Arabic, 0.85 s.d. for French, and 0.60 s.d. for mathematics. Again, all of these estimates are highly significant and are very large effects.

Table 2: *Students' Performance After 3-Year Rollout*

	Cohort 1 (2023/24)				Cohort 2 (2024/25)				Cohort 3 (2025/26)				Cohorts Pooled			
	Overall	Arabic	French	Math	Overall	Arabic	French	Math	Overall	Arabic	French	Math	Overall	Arabic	French	Math
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
Panel A: 1-Year Exposure																
All students	0.903***	0.489***	1.244***	0.987***	0.505***	0.304**	0.755***	0.488***	0.403***	0.250***	0.466***	0.497***	0.604***	0.348***	0.811***	0.656***
	(0.030)	(0.044)	(0.050)	(0.039)	(0.061)	(0.114)	(0.079)	(0.075)	(0.049)	(0.046)	(0.083)	(0.041)	(0.040)	(0.041)	(0.057)	(0.034)
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.012]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
Number of Students	20,784	6,860	6,858	7,066	4,052	1,302	1,356	1,395	11,528	3,908	3,895	3,767	36,363	12,042	12,089	12,239
Number of Schools	276	270	266	268	62	61	62	62	145	145	145	145	483	475	473	475
Panel B: 2-Year Exposure																
All students	0.623***	0.356***	0.953***	0.610***	0.579***	0.305**	0.729***	0.583***					0.606***	0.358***	0.846***	0.601***
	(0.044)	(0.067)	(0.056)	(0.059)	(0.095)	(0.132)	(0.113)	(0.069)					(0.051)	(0.079)	(0.062)	(0.047)
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.034]	[0.000]	[0.000]					[0.000]	[0.000]	[0.000]	[0.000]
Number of Students	11,150	3,726	3,695	3,729	1,631	505	546	573					12,755	4,203	4,221	4,277
Number of Schools	184	182	178	180	33	32	33	33					217	213	211	213
Panel C: 3-Year Exposure																
All students	0.532***	0.238*	0.941***	0.544***												
	(0.084)	(0.136)	(0.099)	(0.124)												
	[0.000]	[0.086]	[0.000]	[0.000]												
Number of Students	4,053	1,324	1,338	1,367												
Number of Schools	88	87	86	87												

Note: This table reports the program's intent-to-treat (ITT) effects among all students after one year of exposure to PSP (Panel A), two years of exposure (Panel B), and three years of exposure (Panel C). Test performance is measured as the standardized raw total test score, where standardization is performed within each exam wave relative to the control group for the corresponding cohort. Each subject level exams (Arabic, French, and Math) is reported alongside a composite score stacking the three sub-samples. Standard errors are clustered at the matched-pair level and shown in parentheses. P-values are shown in brackets. Pooled estimates give each cohort equal total weight. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

5.1.1 Heterogeneity Analysis

Gender: Boys vs. Girls Table 3 presents the same estimates as in Table 1, but separately for the female and male students in our sample. Overall, when combining the three subjects, the estimated impacts are larger for girls than for boys, but the differences are not particularly large. By subject, the impacts on all three subjects (Arabic, French and mathematics) are almost always larger for girls than for boys, the main exception being that while the impacts on French skills for Cohort 2 are slightly higher for boys than for girls. But none of these differences is very large. We conclude that the PSP program is similarly effective at increasing the academic skills of boys and girls.

Table 3: *Heterogeneity in Students' Performance by Gender*

	Cohort 1 (2023/24)				Cohort 2 (2024/25)				Cohort 3 (2025/26)				Cohorts Pooled			
	Overall (1)	Arabic (2)	French (3)	Math (4)	Overall (5)	Arabic (6)	French (7)	Math (8)	Overall (9)	Arabic (10)	French (11)	Math (12)	Overall (13)	Arabic (14)	French (15)	Math (16)
Panel A: 1-Year Exposure																
Female	0.939***	0.475***	1.280***	1.024***	0.538***	0.408***	0.740***	0.537***	0.448***	0.272***	0.509***	0.515***	0.642***	0.383***	0.835***	0.692***
	(0.032)	(0.045)	(0.051)	(0.044)	(0.072)	(0.116)	(0.084)	(0.083)	(0.041)	(0.053)	(0.069)	(0.048)	(0.032)	(0.043)	(0.044)	(0.039)
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.001]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
Number of Students	10,271	3,345	3,420	3,388	1,990	616	677	682	5,559	1,834	1,858	1,805	17,820	5,795	5,955	5,875
Number of Schools	276	270	265	268	62	60	62	62	145	145	145	145	483	475	472	475
Male	0.865***	0.500***	1.185***	0.942***	0.478***	0.224*	0.753***	0.446***	0.362***	0.215***	0.433***	0.441***	0.567***	0.307***	0.775***	0.612***
	(0.033)	(0.050)	(0.060)	(0.047)	(0.061)	(0.119)	(0.082)	(0.072)	(0.050)	(0.046)	(0.084)	(0.044)	(0.041)	(0.045)	(0.061)	(0.036)
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.071]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
Number of Students	10,511	3,437	3,356	3,605	2,062	671	666	710	5,966	2,013	1,987	1,930	18,539	6,121	6,009	6,245
Number of Schools	276	270	265	268	62	60	62	62	145	145	145	145	483	475	472	475
Panel B: 2-Year Exposure																
Female	0.687***	0.370***	1.000***	0.627***	0.672***	0.452***	0.712***	0.634***					0.679***	0.465***	0.859***	0.635***
	(0.047)	(0.071)	(0.065)	(0.069)	(0.115)	(0.125)	(0.120)	(0.103)					(0.059)	(0.075)	(0.070)	(0.063)
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.003]	[0.000]	[0.000]					[0.000]	[0.000]	[0.000]	[0.000]
Number of Students	5,497	1,794	1,869	1,747	781	214	270	272					6,278	2,008	2,139	2,019
Number of Schools	184	182	178	180	33	31	33	33					217	213	211	213
Male	0.561***	0.305***	0.875***	0.607***	0.527***	0.164	0.720***	0.526***					0.547***	0.224**	0.801***	0.573***
	(0.046)	(0.078)	(0.061)	(0.069)	(0.089)	(0.143)	(0.132)	(0.063)					(0.050)	(0.087)	(0.072)	(0.051)
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.267]	[0.000]	[0.000]					[0.000]	[0.011]	[0.000]	[0.000]
Number of Students	5,623	1,863	1,760	1,894	831	263	260	285					6,454	2,126	2,020	2,179
Number of Schools	184	181	178	180	33	31	33	33					217	212	211	213
Panel C: 3-Year Exposure																
Female	0.616***	0.334**	1.039***	0.575***												
	(0.091)	(0.161)	(0.124)	(0.093)												
	[0.000]	[0.044]	[0.000]	[0.000]												
Number of Students	1,970	603	669	627												
Number of Schools	88	87	86	86												
Male	0.465***	0.126	0.808***	0.443***												
	(0.089)	(0.111)	(0.081)	(0.146)												
	[0.000]	[0.261]	[0.000]	[0.004]												
Number of Students	2,032	648	607	682												
Number of Schools	88	87	85	86												

Note: This table reports on the program's intent-to-treat (ITT) effects between female and male students after 1-year of exposure to PSP (Panel A), 2-years of exposure to PSP (Panel B), and 3-years of exposure to PSP (Panel C). The outcome of student performance is measured as the standardized test performance in each exam wave relative to the control group for each sample cohort. Each subject level exams (Arabic, French, and Math) is reported alongside a composite score stacking the three sub-samples. Standard errors are clustered at the matched-pair level and shown in parentheses. P-values are reported in brackets. Pooled estimates are based on cohort-reweighted samples. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Baseline Test Performance Table 4 also presents the same estimates as in Table 1, but it presents separate results for the bottom quartile and top quartile of test scores as measured at baseline. This is done to see whether the program is effective at raising students' skills at both tails of the distribution of initial performance. We ranked students in terms of their baseline performance and separated them into quartiles along this dimension. Then we estimate the impact of the PSP separately for the top quartile of baseline performing students and the bottom quartile of baseline performing students.

After one year of exposure, Cohort 1 students in the bottom quartile increase their performance, averaged over all three tests, by 0.85 s.d. For the top quartile the increase is slightly smaller, at 0.82 s.d. After two years of exposure for Cohort 1 students, the impact on the bottom quartile is – as expected given the results in Table 1 – much smaller, at 0.59 s.d., while the impact on the top quartile also declines, to 0.54 s.d. Finally, after three years of exposure, the impacts drop further for both quartiles: 0.52 s.d. for the bottom quartile and 0.46 s.d. for the top quartile. Overall, the PSP increased student performance for students in both the top and bottom quartiles, but the impact is slightly larger for the bottom quartile of students.

Turning to Cohort 2, and again focusing on the overall scores that average over the three subjects, after one year of exposure the bottom quartile's scores increase by 0.53 s.d. while the top quartile's increase is noticeably smaller (although still large), at 0.37 s.d. After a second year of exposure, the bottom quartile's increase is larger, at 0.63 s.d., while the top quartile's performance is much higher, at 0.50 s.d. Finally, for Cohort 3 after one year of exposure, the overall impact of the PSP is an increase of 0.36 s.d. for the bottom cohort and 0.34 s.d for the top cohort. These results again show that the PSP increased students' skills for both cohorts, but slightly more for the bottom cohort than for the top cohort.

Table 4: *Heterogeneity in Students' Performance by Baseline Performance*

	Cohort 1 (2023/24)				Cohort 2 (2024/25)				Cohort 3 (2025/26)				Cohorts Pooled			
	Overall (1)	Arabic (2)	French (3)	Math (4)	Overall (5)	Arabic (6)	French (7)	Math (8)	Overall (9)	Arabic (10)	French (11)	Math (12)	Overall (13)	Arabic (14)	French (15)	Math (16)
Panel A: 1-Year Exposure																
Bottom quartile	0.854***	0.565***	1.148***	0.872***	0.529***	0.373***	0.577***	0.436***	0.359***	0.213***	0.395***	0.398***	0.595***	0.398***	0.734***	0.574***
	(0.047)	(0.064)	(0.079)	(0.057)	(0.093)	(0.134)	(0.125)	(0.110)	(0.073)	(0.069)	(0.117)	(0.070)	(0.058)	(0.055)	(0.088)	(0.051)
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.009]	[0.000]	[0.000]	[0.000]	[0.003]	[0.001]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
Number of Students	6,573	1,914	2,269	2,200	1,162	323	397	418	3,161	951	1,011	967	10,896	3,177	3,665	3,596
Number of Schools	276	250	249	265	62	56	57	59	145	135	132	134	483	441	438	458
Top quartile	0.816***	0.404***	0.671***	1.245***	0.373***	0.322**	0.639***	0.457***	0.339***	0.168**	0.434***	0.324***	0.469***	0.275***	0.536***	0.567***
	(0.047)	(0.082)	(0.123)	(0.092)	(0.084)	(0.150)	(0.098)	(0.135)	(0.049)	(0.071)	(0.089)	(0.079)	(0.040)	(0.065)	(0.061)	(0.072)
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.041]	[0.000]	[0.002]	[0.000]	[0.020]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
Number of Students	2,980	959	532	649	783	260	192	224	2,386	805	640	602	6,149	2,024	1,364	1,476
Number of Schools	271	213	138	189	61	51	48	54	144	131	110	126	476	395	296	369
Panel B: 2-Year Exposure																
Bottom quartile	0.590***	0.372***	0.796***	0.607***	0.631***	0.304**	0.568**	0.729***					0.618***	0.348***	0.691***	0.667***
	(0.059)	(0.083)	(0.101)	(0.066)	(0.154)	(0.126)	(0.229)	(0.174)					(0.078)	(0.071)	(0.113)	(0.096)
	[0.000]	[0.000]	[0.000]	[0.000]	[0.001]	[0.033]	[0.025]	[0.001]					[0.000]	[0.000]	[0.000]	[0.000]
Number of Students	3,258	1,039	1,178	1,086	468	120	139	149					3,726	1,159	1,317	1,235
Number of Schools	183	169	167	173	33	25	29	28					216	194	196	201
Top quartile	0.542***	0.307**	0.790***	0.709***	0.496***	0.224	0.767***	0.147					0.519***	0.313**	0.773***	0.395***
	(0.054)	(0.122)	(0.136)	(0.141)	(0.128)	(0.338)	(0.202)	(0.191)					(0.071)	(0.143)	(0.138)	(0.120)
	[0.000]	[0.014]	[0.000]	[0.000]	[0.001]	[0.525]	[0.002]	[0.458]					[0.000]	[0.031]	[0.000]	[0.001]
Number of Students	1,855	503	383	450	292	71	94	77					2,147	574	477	527
Number of Schools	182	136	106	133	33	19	27	20					215	155	133	153
Panel C: 3-Year Exposure																
Bottom quartile	0.517***	0.222**	0.770***	0.589***												
	(0.119)	(0.110)	(0.159)	(0.156)												
	[0.000]	[0.050]	[0.000]	[0.000]												
Number of Students	1,251	392	376	438												
Number of Schools	88	79	78	85												
Top quartile	0.463***	0.336**	0.779***	0.370												
	(0.144)	(0.148)	(0.188)	(0.261)												
	[0.002]	[0.029]	[0.000]	[0.166]												
Number of Students	781	203	195	159												
Number of Schools	88	64	54	53												

Note: This table reports on the program's intent-to-treat (ITT) effects between the top quartile and bottom quartile of students based on their baseline test performance after 1-year of exposure to PSP (Panel A), 2-years of exposure to PSP (Panel B), and 3-years of exposure to PSP (Panel C). The outcome of student performance is measured as the standardized test performance in each exam wave relative to the control group for each sample cohort. Each subject level exams (Arabic, French, and Math) is reported alongside a composite score stacking the three sub-samples. Standard errors are clustered at the matched-pair level and shown in parentheses. P-values are reported in brackets. Pooled estimates are based on cohort-reweighted samples. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

6 Understanding Constraints to Scaling

6.1 The Failure of Interventions to Scale

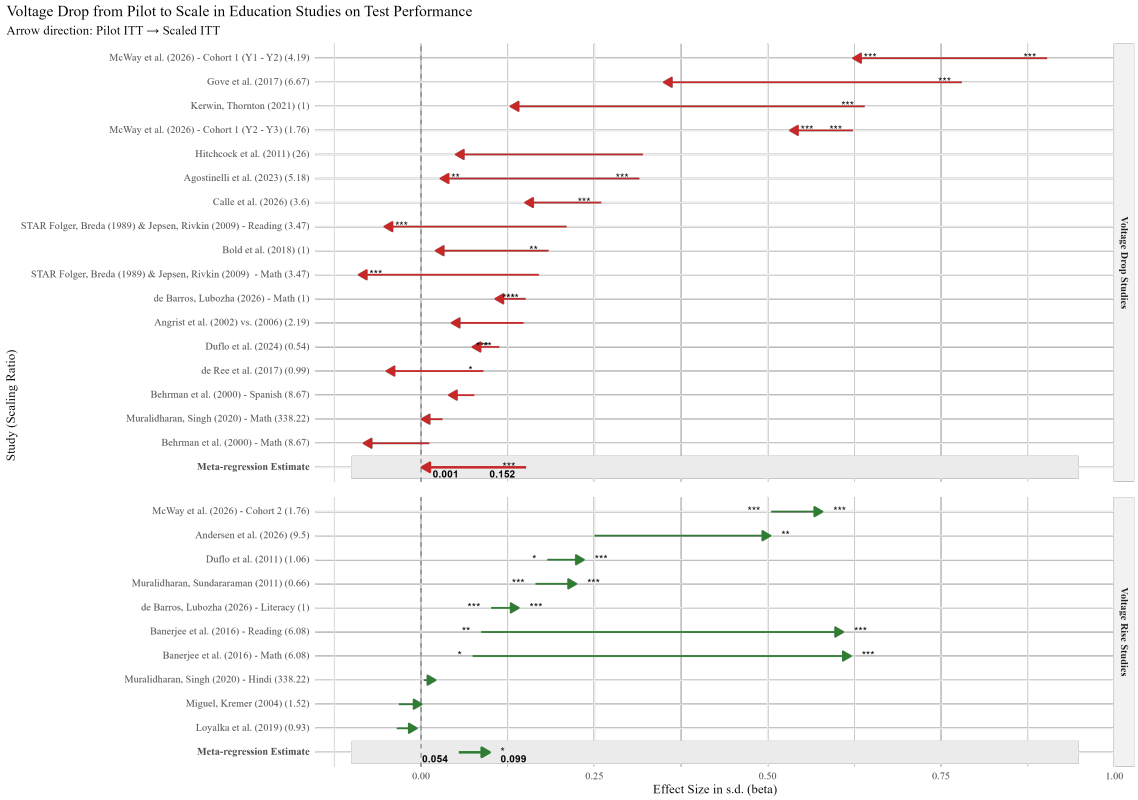
One of the central challenges in the education literature is that interventions that generate large effects in pilot settings often produce much smaller gains when governments attempt to implement them at scale. More broadly, evidence on external validity suggests that treatment effects frequently attenuate as programs move from tightly controlled study settings to routine implementation contexts (Allcott, 2015; Vivalt, 2020). In education, this concern is especially important because many of the most promising interventions rely on a high degree of implementation fidelity, sustained pedagogical support, and organizational capacity that may be difficult to reproduce within government systems.

This issue is particularly relevant for the two "great buy" strategies that motivate the PSP. Targeting instruction to a child's learning level has been shown to generate substantial learning gains, but Banerjee et al. (2017) emphasize that moving from proof of concept to scalable policy requires adaptation to administrative constraints, financing realities, and the limits of state capacity. More recent large-scale evidence likewise shows that some learning interventions remain effective under government delivery, but often with smaller and more uneven impacts than in earlier pilots (Duflo et al., 2024). The same pattern appears for structured pedagogy and scripted lessons: while these approaches can improve learning, their effectiveness depends heavily on whether teachers receive usable materials, regular supervision, and support for consistent classroom implementation (Kerwin and Thornton, 2021; Piper and Dubeck, 2024).

Taken together, this literature suggests that scaling failure is not best understood as evidence that an intervention "does not work." Rather, it often reflects a voltage drop between pilot and scale that arises when governments must deliver complex reforms through existing bureaucracies, with heterogeneous local capacity, weaker monitoring, and less intensive support than in smaller efficacy trials (Allcott, 2015; Vivalt, 2020; Banerjee et al., 2017). This concern is directly relevant for understanding the PSP: the program combines interventions with strong evidence in the literature, but it does so through Morocco's public education system and at a scale where implementation constraints are likely to matter.

Figure 4 displays a review of the literature on voltage drop in education interventions. The figure presents a dumbbell plot of the difference between piloted impact evaluation ITT effect sizes compared to the scaled ITT effect size for education studies measuring student test performance. The top plot facet shows studies with voltage drop and the bottom facet shows studies with voltage rise. The direction of the arrows indicate the change from the pilot to the scale effect. The figure shows that this is a divergence between education studies that see voltage drops versus voltage rises. The success of scaling a pilot intervention is therefore likely dependent on several characteristics which do not always lead to disapation of the results.

Figure 4: *Voltage Drop from Pilot to Scale in Education Studies*



Note: This figure presents a dumbbell plot of the difference between piloted impact evaluation ITT effect sizes compared to the scaled ITT effect size for education studies measuring student test performance. The top plot facet shows studies with voltage drop and the bottom facet shows studies with voltage rise. The direction of the arrows indicate the change from the pilot to the scale effect. Effects are presented in standard deviation changes. Each arrow is labeled by the study as well as the scaling ratio of the pilot sample to the scale sample size in parentheses. Stars signify statistical significance of the pilot and scale estimates: $* < 0.1, ** < 0.05, *** < 0.01$.

6.2 The Voltage Effect (List, 2026): A Model of Scaling

List (2026) presents the 'volatage effect' model. This structural model provides a means of understanding the components underlying the changes program implementaiton that shifts the impact on student learning between pilot and scale. Here I briefly re-describe the core of the model for application in this setting.

Environment and Timing. Consider a simplified lifecycle with two childhood periods followed by adulthood. Period 0 corresponds to early childhood (ages 0-5), period 1 to later childhood and adolescence (ages 6-18), and period 2 to adult labor market outcomes. This stark periodization abstracts from continuous development while capturing the key insight that early investments affect both immediate skills and the productivity of future investments. The population consists of a continuum of children indexed by $i \in [0, 1]$, each with initial skill endowment $\theta_{0,i}$ drawn from distribution F_0 with density f_0, mean μ_0, and variance σ_0^2. For notational convenience, I integrate over the skill distribution using measure $f_0(\theta_{0,i})d\theta_{0,i}$. This initial endowment captures all pre-intervention factors affecting child development: parental education and income, neighborhood quality, prenatal health, genetic endowments, and early home environment quality. Disadvantaged children—those from low-income families, unstable households, or under-resourced communities—are characterized by lower realizations of $\theta_{0,i}$.

Technology of Skill Formation. Skills evolve according to a production function that embeds two core mechanisms from Heckman's framework: self-productivity and dynamic complementarity. The skill production function is given by:

$$\theta_{0,i} = A \cdot \theta_{t-1,i}^{\beta} \cdot (1 + I_{t-1,i})^{\delta} + \varepsilon_{t,i} \tag{2}$$

where $t \in 1, 2$ indexes childhood periods and i indexes individual children, with θ_0 representing initial endowment, θ_1 skills after early childhood (period 0 to 1), and θ_2 skills entering adulthood (period 1 to 2). In what follows, I suppress individual subscripts when describing the production technology generically, reintroducing them when analyzing population distributions. In equation (1), each parameter has a clear economic interpretation: **Self-productivity** ($\in (0, 1)$). The term $\theta_{t-1,i}^{\beta}$ captures "skill begets skill"— children with higher skills in period $t1$ develop more skills in period t. The parameter $A > 0$ governs the overall productivity of the skill formation process, while the exponent $\beta < 1$ implies diminishing returns to existing skills. **Investment technology** ($\delta(0, 1)$). The term $(1 + I_{t-1,i})^{\delta}$ represents how program investments translate into skill gains, where $I_{t,i}ge0$denotes investment intensity measured in resource units (e.g., thousands of dollars per child annually). We use $1 + I$ rather than I to ensure the function is well-defined when $I = 0$. The exponent $\delta < 1$ implies diminishing marginal returns to investment intensity. **Dynamic complementarity**. Critically, the multiplicative form delivers:

$$\frac{\partial^2 \theta_t}{\partial \theta_{t-1}} \partial I_{t-1} = A\beta\delta \cdot \theta_{t-1}^{\beta-1} \cdot (1 + I_{t-1})^{\delta-1} > 0 \tag{3}$$

This positive cross-partial formalizes dynamic complementarity: investments are more productive for children with stronger baseline skills. The marginal product of investment increases with existing skill levels:

$$\frac{\partial}{\partial \theta_{t-1}}[\frac{\partial \theta_t}{\partial I_{t-1}}] = A\beta\delta \cdot \theta_{t-1}^{\beta-1} \cdot (1 + I_{t-1})^{\delta-1} > 0 \tag{4}$$

Equation (3) restates (2) with the order of differentiation reversed, underscoring the symmetry in how prior skills enhance the productivity of investments and vice versa, a property that follows from the commutativity of mixed partial derivatives in the skill production function. Dynamic complementarity creates a tension between equity (targeting low-θ_0 children who are most disadvantaged) and efficiency (investing where marginal returns are highest). The resolution of this tension depends on the degree of diminishing returns (β) and the social planner's welfare function. **Stochastic shocks** ($\varepsilon_{t,i}$). The shock $\varepsilon_{t,i} \sim N(0, \sigma_\varepsilon^2)$ captures idiosyncratic factors affecting development—health shocks, family disruptions, peer influences, or measurement error in skills. These shocks introduce uncertainty but, being meanzero and independent across children, wash out in aggregate welfare calculations. Investment costs. Each unit of investment costs $c_t > 0$, representing the opportunity cost of resources. In the baseline model, c_t is constant (implying perfect supply elasticity). Below, I relax this assumption by introducing $c_t(q)$ with $c_t(q) > 0$ to capture supply constraints that raise marginal costs at scale.

From Pilot to Scale. Consider a targeted pilot conducted on n disadvantaged children with $\theta_{0,1} < \bar{\theta}$. Treated children receive investment $I_0 = I$ in period 0, while controls receive $I_0 = 0$. No subsequent investments occur ($I_1 = 0$), allowing observation of dynamic effects through selfproductivity. From the multiplicative production function (1), early investment generates treatment effects:

$$\Delta\theta_{2,i} = (A^{pilot})^{1+\beta}\theta_{0,i}^{\beta^2}[(1 + I^*)^{\delta\beta} - 1] \tag{5}$$

Two features merit emphasis. First, complementarity generates heterogeneous treatment effects: absolute gains increase with baseline skills $\theta_{0,i}$. The proportional gain in period-1 skills is constant across children—each experiences $(1 + I)^{\delta} 1$ proportional increase in θ_1. However, because these gains compound through self-productivity in period 2, the absolute adult skill gains $\Delta\theta_2$ increase with θ_0. Second, dynamic amplification through self-productivity magnifies early advantages, with the exponent β^2 determining how much baseline heterogeneity affects adult outcomes.

The pilot's average treatment effect is

$$ATE = (A^{pilot})^{1+\beta}[(1 + I^*)^{\delta\beta} - 1] \cdot E[\theta_{0,i}^{\beta^2} | \theta_{0,i} < \bar{\theta}] \tag{6}$$

7 Conclusion

This paper evaluates whether a government can deliver large learning gains while rapidly scaling a national education reform. Using matched-pair difference-in-differences estimates for Morocco's Pioneer School Program, we find that PSP substantially improves student

Table 5: *Sources of Voltage Drops in Early Childhood Interventions*

Mechanism	**Formal Parameter**	**Description**
Quality voltage	$A_{\text{scale}} = (1 - v_A)A_{\text{pilot}}$	Exceptional pilot implementation (master's teachers, 4:1 ratios, intensive oversight) exceeds sustainable scaled quality. Benefits overstate by $(1 - v_A)^{-(1+\beta)}$; with $\beta = 0.7$, $v_A = 0.3$, this equals 2.1×.
Sample composition	$E[\theta_{0,i}^{\beta^2} \mid \text{pilot}] > E[\theta_{0,i}^{\beta^2} \mid \text{scale}]$	Volunteer families within disadvantaged populations have higher baseline skills than automatic enrollment reaches due to selection on parental motivation and organization.
Compliance voltage	$I_{\text{eff}} < I^*$	Volunteer participants attend more regularly (85% vs. 65%), reducing effective dosage at scale by factor $(I_{\text{eff}}/I^*)^{\delta\beta}$.
General equilibrium	Multiple channels	Supply constraints strain provider capacity; public programs crowd out private investments; peer composition changes; skill price effects emerge population-wide.

achievement in Arabic, French, and mathematics. The magnitudes are large in absolute terms and remain high relative to global benchmarks from the education impact literature.

At the same time, the results show that scaling is dynamic rather than mechanical. Effects vary across rollout cohorts and by years of exposure, indicating that implementation context matters for persistence and magnitude. These patterns are consistent with a broader scaling view in which policy effectiveness depends not only on intervention content, but also on delivery quality, supervisory intensity, incentive salience, and operational capacity as reforms expand.

Methodologically, the paper demonstrates how administrative "big data" and field assessments can be combined to evaluate non-random national reforms with credible counterfactuals. High-dimensional matching and post-double-selection controls improve comparability at baseline, while pre-trend diagnostics and stable-sample checks increase confidence that core findings are not driven by compositional change.

From a policy perspective, the evidence suggests that system-level reforms centered on structured pedagogy and targeted remediation can generate meaningful human-capital gains under public delivery. The key implication is not that scaling inevitably weakens impact, but that maintaining impact at scale requires institutional attention to the mechanisms that govern implementation. Ongoing work with the voltage-effect framework formalizes these mechanisms and provides a basis for counterfactual analysis of how alternative implementation choices may affect long-run learning outcomes in Morocco and in other low-performing education systems.

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Supplementary Appendix

A PSP School Adoption Requirements

Participation in the Pioneer School Program (PSP) is governed by a formal conditional commitment mechanism designed to ensure institutional ownership, pedagogical alignment, and measurable accountability.

A.1 Contractual Adhesion and Governance Requirements

Voluntary Sign-off To obtain PSP status, the school director and a minimum of 70% of the teaching staff must sign a formal written commitment to:

- Adopt the new pedagogical approaches (Teaching at the Right Level and Explicit Teaching),
- Participate in the required training sessions,
- Undergo individual certification processes,
- Comply with monitoring and reporting requirements.

PSP participation is therefore predicated on a bottom-up adhesion mechanism, intended to signal collective engagement at the school level.

Projet d'Établissement Intégré (PEI) Each participating school must adopt and implement a Projet d'Établissement Intégré (PEI), approved by the Provincial Directorate.

Unlike traditional school development plans, the PSP PEI functions as a performance contract aligning school-level action plans with the National Roadmap 2022–2026. It must include quantified and time-bound targets for:

- Improvement in learning mastery (based on standardized assessments),
- Reduction of dropout rates (statutory objective of approximately 30% reduction),
- Extracurricular participation and student engagement hours.

A.2 Operational and Pedagogical Implementation Requirements

Schools are required to implement the full multi-component intervention bundle. Compliance extends beyond classroom practices to administrative and documentation obligations.

Remediation (TaRL) Schools are required to implement the Teaching at the Right Level (TaRL) protocol as defined by MENPS guidelines. This includes conducting diagnostic assessments at the beginning of the academic year, regrouping students by proficiency level, suspending the regular curriculum during the intensive remediation period (4–6 weeks depending on cohort), and organizing continued support sessions for students who do not reach grade-level proficiency.

Detailed operational procedures are documented in the program implementation section.

Structured Pedagogy (Explicit Teaching) Schools must deliver the core curriculum (Arabic, French, and Mathematics) using the standardized Explicit Teaching framework mandated by MENPS. This includes the compulsory use of ministry-provided scripted lesson plans and digital instructional materials, and adherence to the prescribed instructional sequence.

Technical details on the pedagogical model and its evolution across cohorts are provided in the program components section.

Competency Tracking and Documentation Teachers are required to maintain individual *Livrets de Compétences* (Competency Booklets) for each student. These booklets:

- Track mastery of specific skills (e.g., letter recognition, word reading, subtraction),
- Serve as documentary evidence of progression,
- Constitute primary physical documentation during inspection visits and certification audits.

A.3 Assessment, Data Entry, and Verification Protocols

The PSP imposes a high-frequency performance reporting regime.

Massar Data Entry Schools must systematically enter assessment results into the Massar information system, including:

- Pre-test scores,
- Post-test scores,
- Periodic Paliers (proficiency level) checks.

At the national scale, this represents the processing of approximately 24 million data points every six weeks.

Quality Assurance and Concordance Verification Schools are subject to a verification protocol whereby pedagogical inspectors:

- Randomly sample student test copies,
- Cross-check recorded scores against entries in Massar,
- Calculate a concordance rate between teacher grading and inspector verification,
- Validate or request correction prior to final dashboard consolidation.

A.4 Monitoring, Certification, and Incentive Framework

Compliance is evaluated under the "École Pionnière" Label framework (Decree $n°$ 2.24.144, July 2024).

B Evolution of the Program Over Time

The Pioneer School Program (PSP) has evolved substantially since its launch. Rather than a fixed intervention, it has undergone structural adjustments in scale, governance, instructional intensity, monitoring architecture, and incentive design across successive phases.

B.1 Phase 1 (2023–2024): Experimental Pilot – Cohort 1

The first year constituted a large-scale experimental phase implemented in 626 primary schools across all 12 regions.

This phase was characterized by:

- A high-intensity instructional model, including a 6-week intensive remediation period at the beginning of the academic year;
- A tightly conditional incentive structure linking individual teacher bonuses (10,000 MAD annually) to both school-level labeling and individual certification;
- Direct inspector-led cascade training and close pedagogical supervision;
- Initial logistical constraints, including delays in the delivery of textbooks and digital projection equipment in some regions.

Implementation during this phase coincided with nationwide teacher strikes lasting approximately three months, which disrupted instruction and created regional heterogeneity in exposure and implementation fidelity.

Phase 2 (2024–2025): Rapid Expansion – Cohort 2

The second phase marked a transition from experimental pilot to systemic scaling.

Primary-level coverage expanded from 626 to approximately 2,626 schools, representing roughly 30% of public primary enrollment. The reform simultaneously extended to the lower secondary cycle through the launch of 232 "Collèges Pionniers."

This expansion was accompanied by several structural adjustments:

- A formal reduction in the duration of the intensive remediation period (from 6 to 4 weeks for newly entering schools);
- Strengthening of logistical coordination, with improved availability of instructional materials at the start of the academic year;
- Reconfiguration of the incentive regime following the resolution of the 2023–2024 social unrest.

In particular, the originally PSP-specific conditional performance bonus was largely integrated into a broader nationwide salary adjustment for teachers. As a result, in Year 2 the marginal financial incentive specific to PSP participation became less distinct from the general compensation framework, potentially altering incentive salience and motivation dynamics.

The expansion phase also required scaling of supervisory capacity, with a substantial increase in the number of inspectors and teachers operating under the PSP framework.

Phase 3 (2025–2026 onward): Progressive Generalization

Beginning in 2025–2026, the PSP entered a progressive generalization phase, with the addition of approximately 2,000 primary schools per year.

The medium-term objective is to reach approximately 50% coverage of public primary establishments by 2025–2026, with full national generalization (approximately 8,630 establishments) projected for the 2027–2028 academic year.

C Test Exam (Instrument) Design

C.1 Item Selection

We administer four exam surveys to measure student performance over three years. Two of these are baseline exams to measure baseline performance separately for Cohort 1 and Cohort 2. The remaining two exams are endline exams to measure the progression in student test performance at the end of the first and second year of the evaluation period. Table 6 presents the number of question items for each exam administered at each grade level across the three subjects. Test items were constructed in coordination with the Ministry of Education.

Examines were administered by enumeration staff provided by the Ministry of Education with permission from school administrations. Notably, in the second year we do not test grade 1 performance.[4] Additionally, the majority of French subject exams were shared across grade levels.

To assess instrument quality, we examined item difficulty and discrimination for each subject-by-grade test form using a two-parameter logistic (2PL) item response theory (IRT) model. "Don't know" responses were scored as incorrect in the main specification, consistent across all four survey exams.

Across subjects, item difficulties covered a broad range, indicating that the tests included both easy and challenging items rather than concentrating at a single difficulty level. In mathematics, difficulty parameters ranged from 2.36 to 2.33 and discrimination parameters from 0.34 to 3.63, confirming that the instrument appropriately captured the proficiency range expected across grades. For Arabic, difficulty ranged from 2.51 to 3.60 and discrimination from 0.30 to 4.23, showing broad coverage of student ability levels with most items providing meaningful differentiation and a small subset of weakly discriminating items. The French test exhibited difficulty parameters between 1.99 and 0.90 and discrimination between 0.70 and 4.83. The estimated item difficulties for French are concentrated at lower ability levels, implying that the French test provides the most measurement precision for below- to average-ability students and less precision in the upper tail.

Overall, item discriminations were positive and of meaningful magnitude across all subjects, indicating that most items differentiated higher- and lower-performing students in the expected direction. Only a small fraction of items displayed weak discrimination or extreme difficulty values, suggesting generally sound psychometric quality.

C.2 Description of Test Scores Over Time

In this section, we provide a description of how test performance is changing between the PSP school students and comparison school students. For each cohort of the study roll out, we compare the distribution of test scores across treatment and control at each exam survey wave. Namely, the baseline and the distribution for each subsequent year of endline exams we evaluate for each cohort. We separate each cohort into their own figure to isolate the variation in change of test performance between treatment and control within comparable groups. In these figures, we denote the control group with dashed lines and the PSP group with solid lines. We denote exam survey waves through color: baseline is red, one year of exposure is blue, two years of exposure is green, and three years of exposure is orange.

Figure 5 shows the distributions for each exam wave for Cohort 1. This was the first cohort in our study and we have followed them over four examination periods with baseline in September 2023. They have received three years of exposure to the PSP program. By design, our matched pair design is confirmed by the close overlap of baseline distributions across treatment status. Then at the end of one year of exposure both groups improve their performance. While expected since the students are learning throughout the academic year, the PSP students gain

[4]The idea with the year 2 evaluation was to follow students from year 1 with Grade 1 students advancing to Grade 2.

Table 6: *Item Map for Question Items by Exam and Grade Level*

	Year 1		Year 2	
	Baseline Exam Cohort 1	Endline Exam Cohort 1	Baseline Exam Cohort 2	Endline Exam Cohort 1 & 2
Arabic				
Grade 1	36 Q (19 A) [262 S]	35 Q (19 A) [241 S]	Untested	Untested
Grade 2	34 Q (9 A) [273 S]	33 Q (18 A) [254 S]	35 Q (9 A) [64 S]	33 Q (9 A) [231 S]
Grade 3	31 Q (14 A) [272 S]	34 Q (23 A) [262 S]	33 Q (13 A) [64 S]	34 Q (16 A) [237 S]
Grade 4	38 Q (15 A) [272 S]	34 Q (21 A) [263 S]	34 Q (16 A) [63 S]	34 Q (14 A) [235 S]
Grade 5	43 Q (6 A) [275 S]	35 Q (13 A) [266 S]	34 Q (14 A) [62 S]	35 Q (16 A) [234 S]
Grade 6	43 Q (17 A) [274 S]	38 Q (31 A) [263 S]	35 Q (17 A) [61 S]	37 Q (12 A) [235 S]
French				
Grade 1	63 Q (26 A) [254 S]	49 Q (18 A) [245 S]	Untested	Untested
Grade 2	72 Q (20 A) [260 S]^{A}	50 Q (32 A) [252 S]^{A}	49 Q (7 A) [66 S]	50 Q (10 A) [224 S]^{A}
Grade 3	72 Q (20 A) [270 S]^{A}	50 Q (32 A) [264 S]^{A}	50 Q (8 A) [65 S]^{C}	50 Q (10 A) [231 S]^{A}
Grade 4	45 Q (7 A) [271 S]^{B}	35 Q (20 A) [264 S]^{B}	50 Q (8 A) [65 S]^{C}	35 Q (10 A) [233 S]^{B}
Grade 5	45 Q (7 A) [268 S] ^{B}	35 Q (20 A) [258 S]^{B}	35 Q (9 A) [64 S]^{D}	35 Q (10 A) [235 S]^{B}
Grade 6	45 Q (7 A) [268 S]^{B}	35 Q (20 A) [260 S]^{B}	35 Q (9 A) [63 S]^{D}	35 Q (10 A) [231 S]^{B}
Mathematics				
Grade 1	38 Q (14 A) [240 S]	26 Q (12 A) [236 S]	Untested	Untested
Grade 2	38 Q (11 A) [266 S]	25 Q (16 A) [263 S]	26 Q (7 A) [67 S]	28 Q (8 A) [222 S]
Grade 3	48 Q (14 A) [266 S]	29 Q (18 A) [264 S]	27 Q (8 A) [63 S]	29 Q (12 A) [237 S]
Grade 4	61 Q (14 A) [270 S]	28 Q (18 A) [268 S]	27 Q (7 A) [67 S]	29 Q (13 A) [235 S]
Grade 5	66 Q (16 A) [269 S]	32 Q (22 A) [267 S]	28 Q (5 A) [65 S]	34 Q (12 A) [238 S]
Grade 6	59 Q (12 A) [269 S]	27 Q (22 A) [265 S]	32 Q (7 A) [64 S]	26 Q (8 A) [237 S]

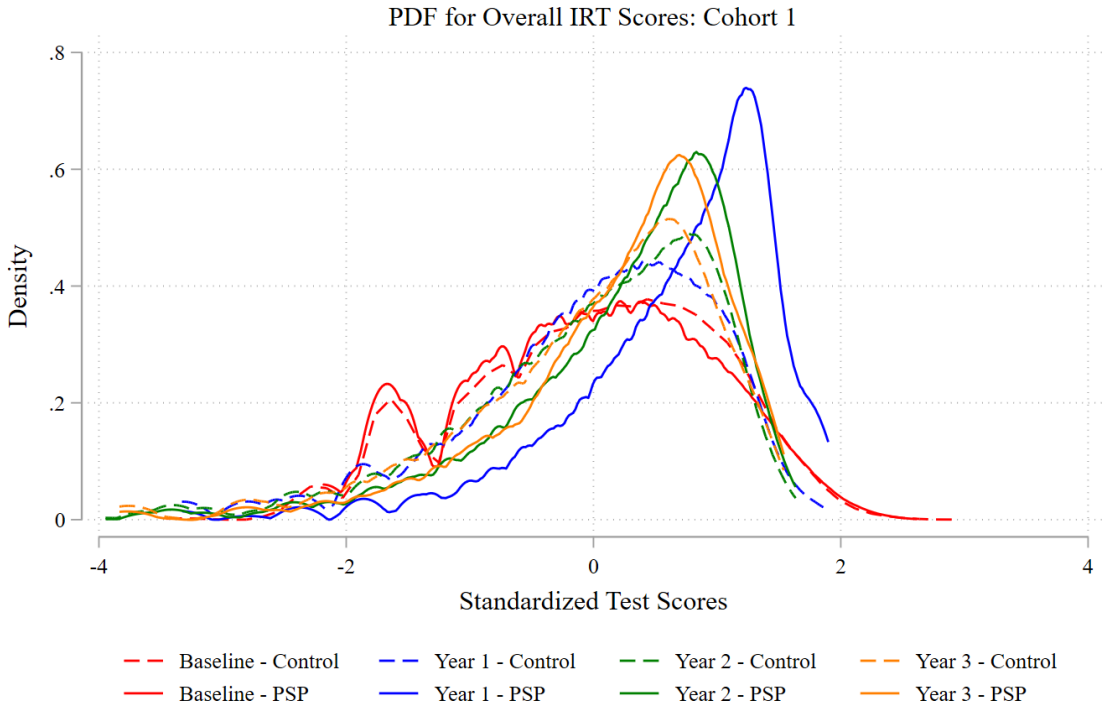
Note: This table presents the number of question items for each exam administered at each grade level across the three subjects. The first number is number of question items (Q) on the exam. In parentheses are the number of anchor item questions (A) on the exam. The brackets includes the number of schools (S) that administered this grade level exam. *A*: Same test for Grade 2 and 3, *B*: Same test for Grade 4, 5 and 6. *C*: Same test for Grade 3 and 4. *D*: Same test for Grade 5 and 6.

a very large increase in their test performance. After two years of exposure, the control group in the lower end of the distribution mirrors the same ability as after 1 year of exposure. But the upper tail of the control distribution has shifted to the right indicating an improvement for top performing students in the control group. By contrast, the entire PSP distribution after two years of exposure has shifted to the left. This suggest that, while performing better than the control students, they students are under-performing compared to their ability after 1 year of exposure. Finally, after three years of exposure both the comparison group and treatment groups have converged with substantial overlap through the ability distribution of test performance. The gains from the PSP program for the lower portion of the distribution of test performance is what drives the results after 3 years of exposure to the PSP program. This is evident since the PSP distribution is entirely to the right of the control distribution below the average performance (i.e., 0). The right portion of the distributions for both PSP and comparison student are nearly identical. This may indicate that while top performing PSP students remain roughly at the same ability from year 2 to year 3, the top performing students in school schools have entirely caught up to the top performing students in PSP schools.

Figure 6 shows the distributions for each exam wave for Cohort 2. This cohort has two years of exposure to the PSP program. Again, our matched pair design is confirmed by the close overlap of baseline distributions across treatment status. After one year of exposure to PSP, PSP students see a substantial improvement in their ability across the whole distribution. By contrast, control students have nearly no improvement in their ability compared to baseline. A way to interpret this is that control students have the same ability between the start and end of the academic school year. After two years of exposure, we begin to see the same phenomenon that we observe with the Cohort 1 test performance. Not only are the top performing PSP students regressing in their ability, but top performing control students are catching up in ability.

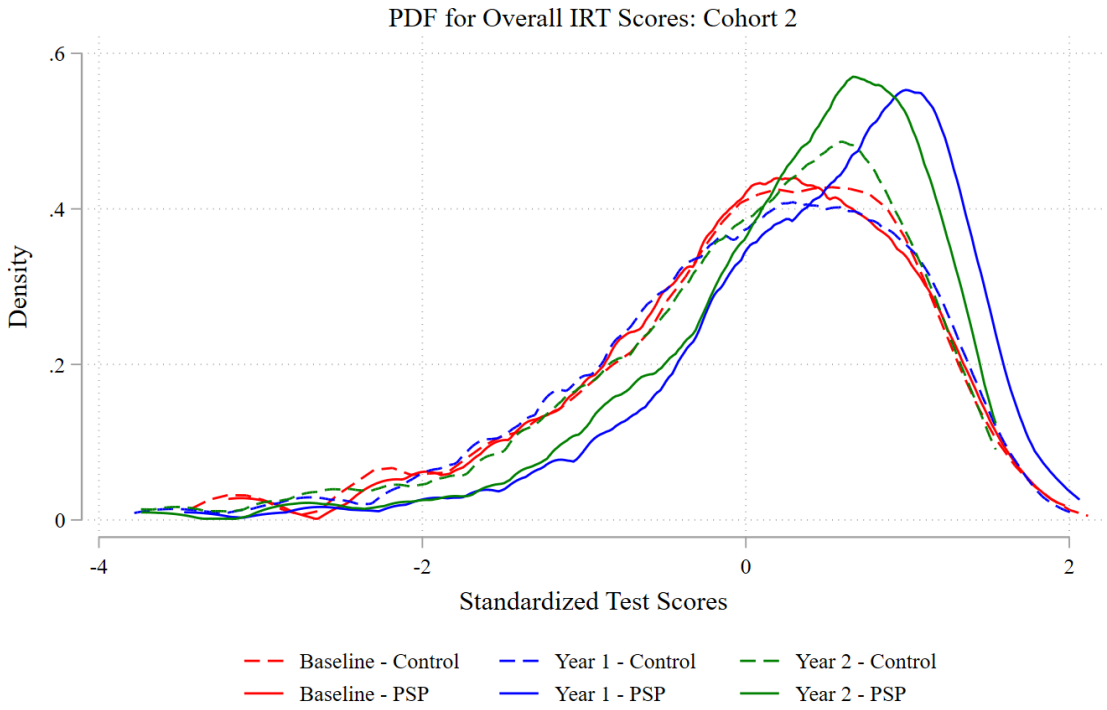
Figure 7 shows the distributions for each exam wave for Cohort 3. This the newest cohort with only one year of exposure to the PSP program. Again, our matched pair design is confirmed by the close overlap of baseline distributions across treatment status. Very similar to the Cohort 2 after one year of exposure, the PSP have improvements along the whole distribution while the control students do not appear to have improved their test performance after a year of academic instruction – regardless of the changes to instruction introduced by the PSP education reform.

Figure 5: *PDF of Test Scores across Survey Waves for Cohort 1*



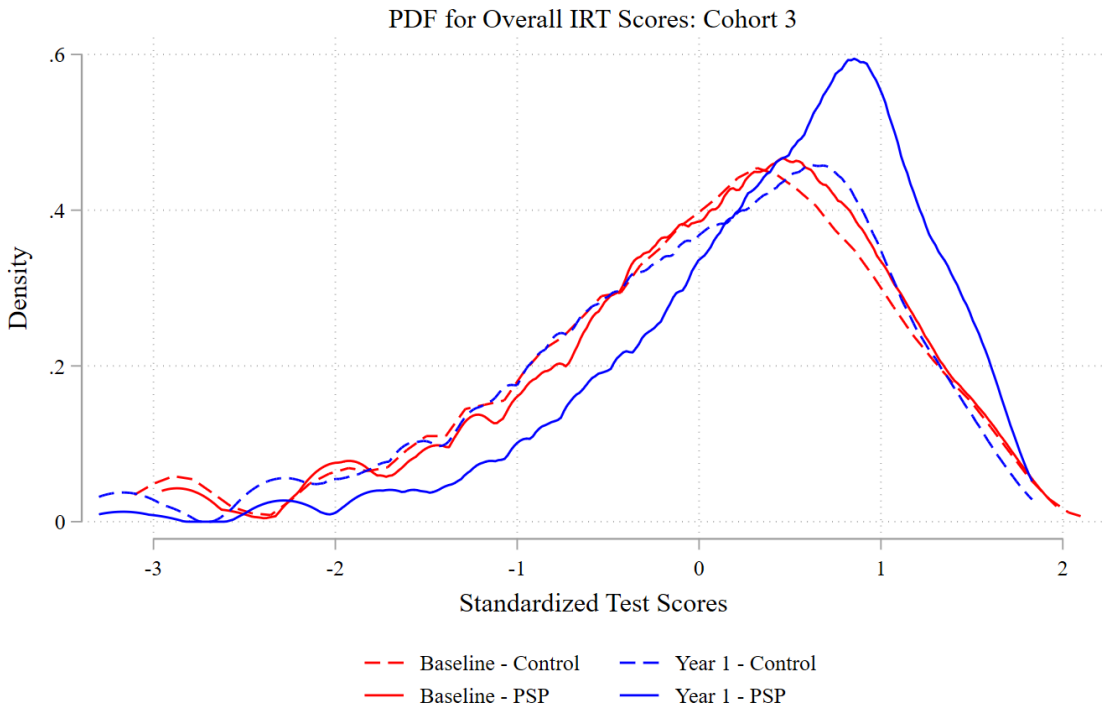
Note: This figure shows the probability distribution function (PDF) for the standardized test performance as a composite overall score (across test subjects in Arabic, French, and mathematics) by the wave of the exam. Red represents the baseline exam. Blue represents the endline exam after 1 year. Green represents the endline exam after 2 years. And orange represents the endline exam after 3 years. The dashed lines represent students in comparison schools, and solid lines represent students in PSP schools.

Figure 6: *PDF of Test Scores across Survey Waves for Cohort 2*



Note: This figure shows the probability distribution function (PDF) for the standardized test performance as a composite overall score (across test subjects in Arabic, French, and mathematics) by the wave of the exam. Red represents the baseline exam. Blue represents the endline exam after 1 year. Green represents the endline exam after 2 years. The dashed lines represent students in comparison schools, and solid lines represent students in PSP schools.

Figure 7: *PDF of Test Scores across Survey Waves for Cohort 3*



Note: This figure shows the probability distribution function (PDF) for the standardized test performance as a composite overall score (across test subjects in Arabic, French, and mathematics) by the wave of the exam. Red represents the baseline exam. Blue represents the endline exam after 1 year. The dashed lines represent students in comparison schools, and solid lines represent students in PSP schools.

D Differential Attrition

D.1 Stable Sample Analysis

The changes over time for the Cohort 1 and 2 results in Table 2 could be misleading because the samples change, becoming smaller from one year to the next because some comparison schools participate in the PSP in later years, and so those schools, and their paired PSP schools, must be dropped from the analysis. To get a clearer understanding of how the estimated impacts change over time, this section presents re-estimated results of the impact of the PSP for Cohorts 1 and 2 using only the schools that are still in the sample in the third year of data collection. This removes potential concerns that the changes in the effectiveness of PSP are due to changes in the composition of the students taking the exam. For example, a drop in the ITT effect may occur because the pairs of schools that are dropped from the sample over time are pairs where the differences in the performance of the PSP and the comparison school were unusually large. This would artificially deflate the estimated ITT effects over time.

We present these results in Table 7.

Table 7: *Students Performance After 3 Years with Stable Samples*

	1-Year Exposure				2-Year Exposure				3-Year Exposure			
	Overall	Arabic	French	Math	Overall	Arabic	French	Math	Overall	Arabic	French	Math
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Panel A: Cohort 1 (2023/24)												
All students	0.964***	0.514***	1.506***	0.888***	0.566***	0.299***	0.880***	0.538***	0.441***	0.167**	0.729***	0.530***
	(0.057)	(0.086)	(0.107)	(0.081)	(0.059)	(0.104)	(0.072)	(0.085)	(0.069)	(0.063)	(0.078)	(0.121)
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.006]	[0.000]	[0.000]	[0.000]	[0.012]	[0.000]	[0.000]
Number of Students	3,711	1,213	1,204	1,294	3,920	1,283	1,299	1,338	3,932	1,286	1,300	1,346
Number of Schools	86	83	83	85	86	85	84	85	86	85	84	86
Panel B: Cohort 2 (2024/25)												
All students	0.514***	0.304*	0.671***	0.435***	0.474***	0.231**	0.552***	0.569***				
	(0.101)	(0.151)	(0.140)	(0.107)	(0.073)	(0.103)	(0.086)	(0.067)				
	[0.000]	[0.064]	[0.000]	[0.001]	[0.000]	[0.041]	[0.000]	[0.000]				
Number of Students	1,486	464	500	522	1,577	488	533	556				
Number of Schools	30	29	30	30	32	31	32	32				

Note: This table reports on the program's intent-to-treat (ITT) effects among all students after for Cohort 1 (Panel A) and Cohort 2 (Panel B). The samples have been modified to become stable to the attrition observed at the end of 3 years of data collection. The outcome of student performance is measured as the standardized test performance in each exam wave relative to the control group for each sample cohort. Each subject level exams (Arabic, French, and Math) is reported alongside a composite score stacking the three sub-samples. Standard errors are clustered at the matched-pair level and shown in parentheses. P-values are reported in brackets. Pooled estimates are based on cohort-reweighted samples. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The results in Table 7 show patterns that are similar to those in Table 1, which indicates that the patterns in Table 1 are not due to changes in the samples of schools and students over time. More specifically, for Cohort 1 there is still a sizable drop in the impact of the PSP after two years of exposure, relative to one year of exposure, and there is a smaller drop after three years of exposure (relative to two years of exposure) than the drop observed after two years of exposure (relative to one year of exposure). For Cohort 2, the impact after two years of exposure is slightly smaller than the impact after one year and thus does not show the

modest rise seen in Table 1 (for all three subjects combined, the increase in Table 1 is from 0.51 s.d to 0.58 s.d., while the decrease in Table 2 is from 0.51 s.d. to 0.47 s.d.) . Yet for two of the three subjects the direction of change is the same: the estimated impact declined in French and increased in mathematics, and only the Arabic result reverses sign (from a tiny increase in Table 1 to a sizable decrease in Table 2). We conclude that, overall, the changes in estimated impacts over time for Cohorts 1 and 2 in Table 1 cannot be explained by changes in the sample over time.

E Internal Validity

E.1 Evidence for Parallel Trends

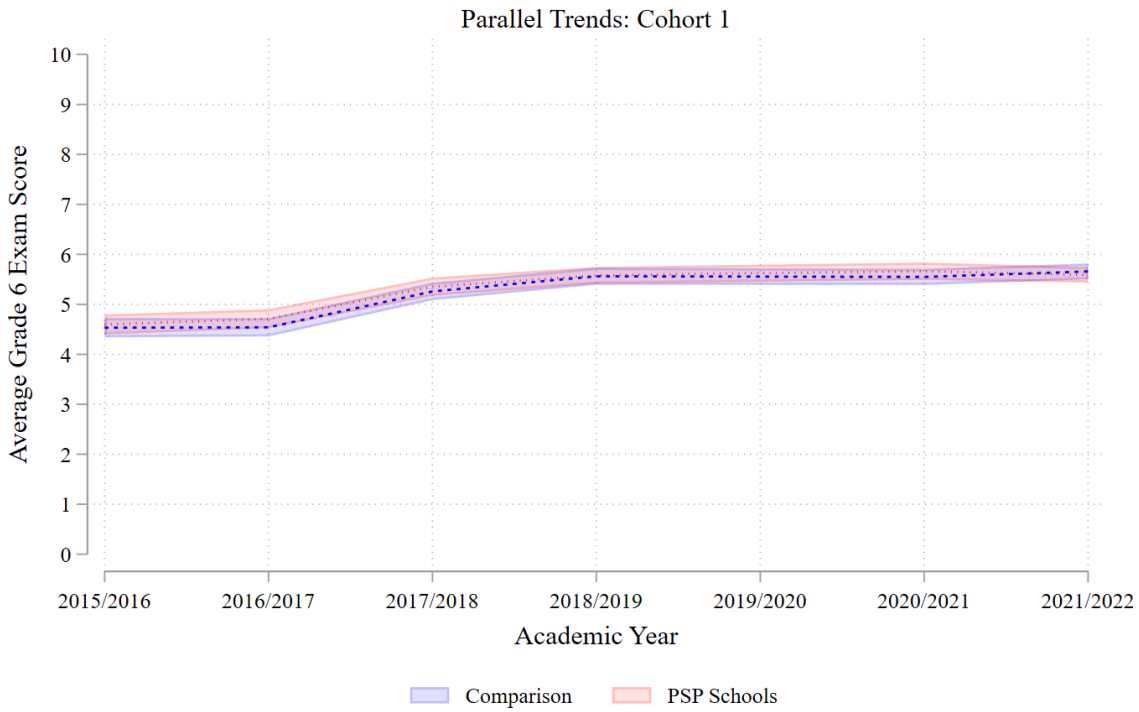
In this section, we report on the internal and external validity of our preliminary estimates. First, we rely on a difference in differences design to recover causal effects. The underlying assumption of this design is the parallel trends assumption. This states that we can interpret our results causally if the trend in the outcome measure for both PSP and comparison schools does not significantly vary before the start of the program. The worry is that anticipation or selection may bias our results.

In the design of our samples, we select schools in matched pairs across PSP and comparison along the school attributes that best predict Grade 6 exam scores. We use this exam score as proxy for the test performance we instrument in our study and report in Table 2. Using 7 years of administrative data, we show below that we assert this assumption for our outcome measure of interest for all three cohorts. Figures 8, 9, and 10 show these trend lines. This provides confidence in the internal validity of our results to interpret our estimates as causal impacts of PSP on student test performance. We also explore the internal validity of each cohort sample by comparing the observable characteristics across PSP and comparison schools in Table 8. By design, these differences are minimal. And where differences do appear, the LASSO PDS correction accounts for these differences in our estimation equation.

E.2 Balance on Observables

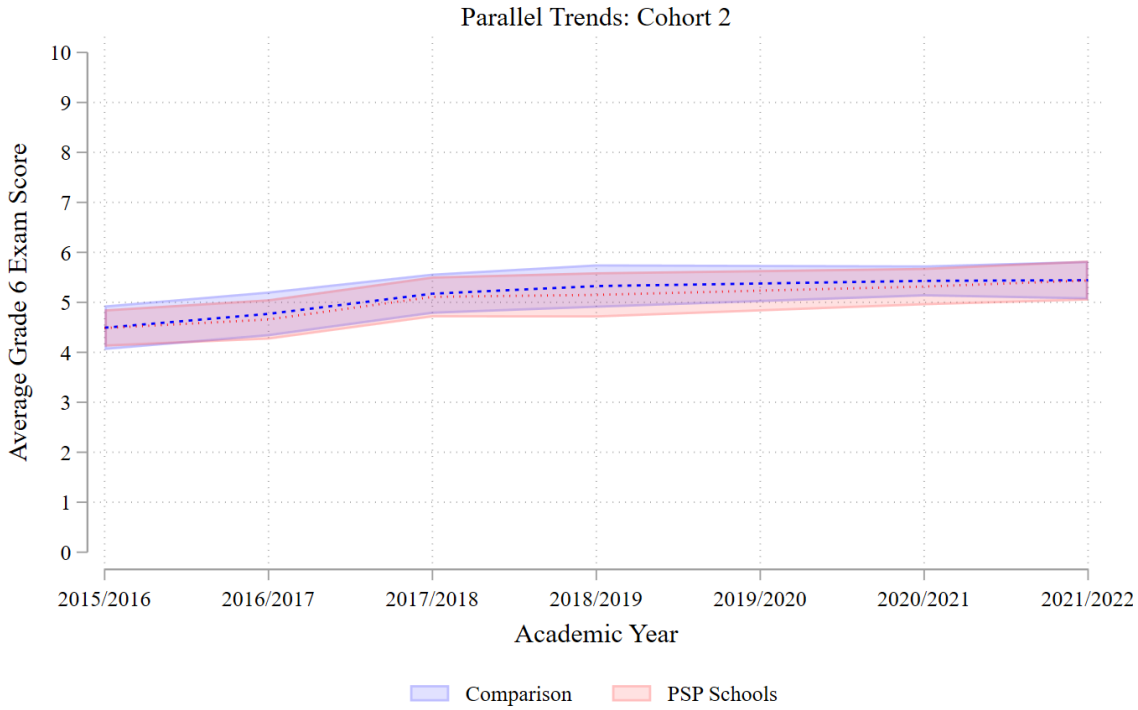
Additionally, we are concerned with whether the schools in this study are similar to the broader set of schools in Morocco. The impact of PSP for the schools in our sample is likely to translate to similar schools. Therefore, we test the observable difference of our study schools compared to schools not selected for our study. Table 9 reports on the external validity of our sample across the three cohorts compared to non-study schools. While there are some considerable differences is the composition of schools in our study compared to non-study schools, importantly the study schools are not different from non-study schools along the dimension of test performance. So we are confident that student ability is not a marked distinction of our study sample. Therefore, the PSP impacts are generalizable to the average student in Morocco.

Figure 8: *Parallel Trends of Test Scores Across Treatment Assignment by Cohort 1 School*



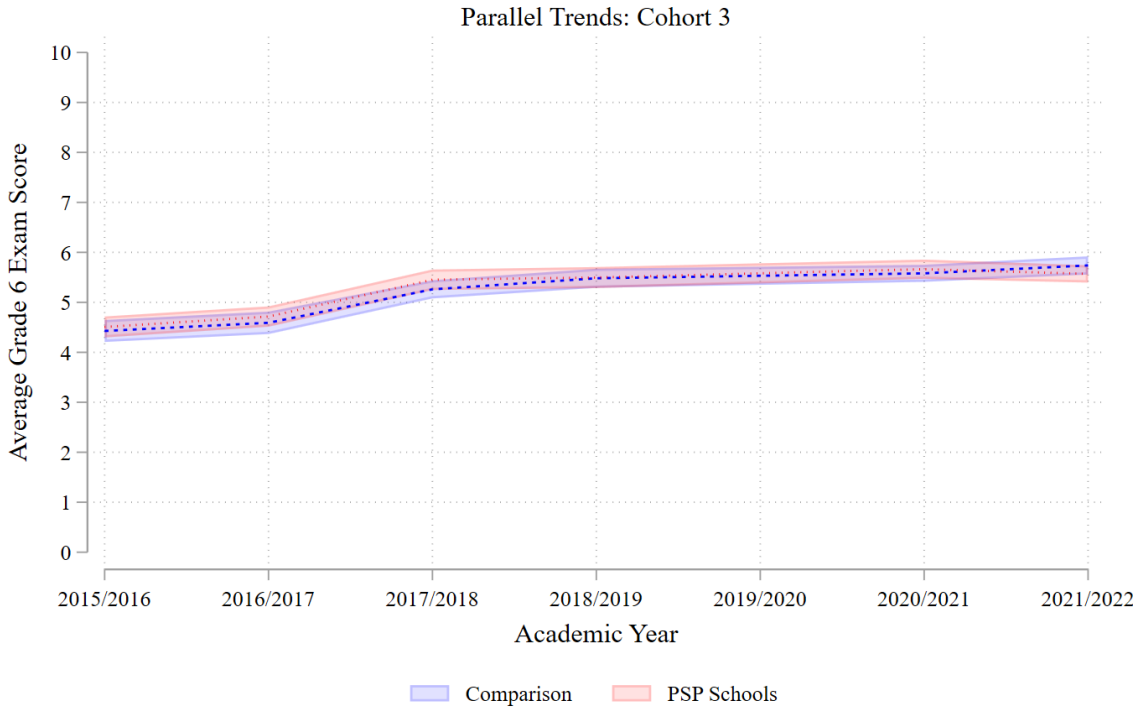
Note: This figure confirms that the program and comparison schools exhibited similar levels and similar trends in exam scores before the program started (2022/23). It presents the yearly average end-of-primary exam scores over a period of seven years before the program launched. Cohort 1 represents the original 300 selected schools (150 PSP, 150 comparison).

Figure 9: *Parallel Trends of Test Scores Across Treatment Assignment by Cohort 2 School*



Note: This figure confirms that the program and comparison schools exhibited similar levels and similar trends in exam scores before the program started (2022/23). It presents the yearly average end-of-primary exam scores over a period of seven years before the program launched. Cohort 2 represents the original 70 selected schools (35 PSP, 35 comparison).

Figure 10: *Parallel Trends of Test Scores Across Treatment Assignment by Cohort 2 School*



Note: This figure confirms that the program and comparison schools exhibited similar levels and similar trends in exam scores before the program started (2022/23). It presents the yearly average end-of-primary exam scores over a period of seven years before the program launched. Cohort 3 represents the original 270 selected schools (135 PSP, 135 comparison).

Table 8: *Internal Validity of Schools in Study Sample*

	Comparison - Cohort 1	PSP - Cohort 1	Difference	Comparison - Cohort 2	PSP - Cohort 2	Difference	Comparison - Cohort 3	PSP - Cohort 3	Difference
Grade 6 Final Exam Score (avg. per classroom, 2021/2022)	5.651	5.595	0.013	5.443	5.436	0.323	5.739	5.569	-0.162
	0.976	0.994	0.899	1.166	1.210	0.237	1.064	1.026	0.171
Grade 6 Final Exam Score (avg. per classroom, 2020/2021)	5.549	5.671	0.173	5.429	5.315	0.158	5.582	5.663	0.077
	0.968	1.140	0.104	0.938	1.134	0.503	1.007	1.143	0.536
Grade 6 Final Exam Score (avg. per classroom, 2018/2019)	5.569	5.575	0.061	5.325	5.149	-0.195	5.482	5.496	0.051
	1.026	1.085	0.574	1.314	1.370	0.498	1.127	1.250	0.707
Grade 6 Final Exam Score (avg. per classroom, 2017/2018)	5.275	5.342	0.174	5.173	5.108	-0.123	5.260	5.452	0.255
	1.053	1.197	0.118	1.216	1.236	0.604	1.079	1.221	0.038
Grade 6 Final Exam Score (avg. per classroom, 2016/2017)	4.564	4.699	0.242	4.769	4.658	-0.012	4.590	4.716	0.197
	1.119	1.226	0.043	1.355	1.221	0.963	1.332	1.204	0.178
Grade 6 Final Exam Score (avg. per classroom, 2015/2016)	4.535	4.601	0.142	4.492	4.486	-0.088	4.427	4.507	0.107
	1.177	1.273	0.286	1.347	1.129	0.724	1.303	1.248	0.454
Total Enrollment (2021/2022)	508.420	497.727	-14.910	497.400	484.114	-28.002	450.704	448.281	-7.237
	332.208	312.852	0.699	326.525	332.736	0.776	370.485	330.399	0.865
Female Enrollment (%, 2021/2022)	0.482	0.483	-0.000	0.485	0.478	-0.009	0.481	0.481	0.001
	0.032	0.034	0.963	0.037	0.029	0.338	0.042	0.032	0.873
Students Repeating (%, 2021/2022)	0.080	0.070	-0.011	0.083	0.086	-0.006	0.085	0.078	-0.006
	0.039	0.033	0.013	0.039	0.041	0.514	0.045	0.038	0.215
Student Dropout (%, 2021/2022)	0.013	0.010	-0.003	0.010	0.009	-0.002	0.011	0.011	0.000
	0.015	0.012	0.042	0.012	0.010	0.582	0.016	0.013	0.971
Students Qualifying for Tayssir (%, 2021/2022)	0.201	0.208	-0.005	0.244	0.217	-0.056	0.370	0.345	-0.030
	0.307	0.290	0.886	0.331	0.291	0.480	0.329	0.327	0.435
Students Qualifying for Social Security (%, 2021/2022)	0.203	0.209	-0.005	0.244	0.217	-0.055	0.372	0.346	-0.030
	0.311	0.291	0.889	0.331	0.292	0.484	0.331	0.327	0.431
Urban	0.680	0.640	-0.024	0.629	0.629	0.060	0.400	0.430	0.030
	0.468	0.482	0.651	0.490	0.490	0.650	0.492	0.497	0.607
Number of Teachers	18.620	18.473	-0.330	18.800	18.743	-0.670	16.711	17.141	0.370
	10.137	10.073	0.789	9.737	9.784	0.818	11.542	10.842	0.788
School's Inspector is Trained	0.820	0.833	0.072	0.171	0.143	-0.033	0.400	0.370	-0.051
	0.385	0.374	0.063	0.382	0.355	0.716	0.492	0.485	0.395
School Type: Community	0.007	0.020	0.019	0.029	0.029	-0.009	0.007	0.022	0.018
	0.082	0.140	0.194	0.169	0.169	0.836	0.086	0.148	0.255
School Type: Central/Sector	0.280	0.287	-0.019	0.286	0.286	-0.091	0.489	0.452	-0.044
	0.451	0.454	0.707	0.458	0.458	0.432	0.502	0.500	0.476
Share of Multi-level classrooms (%, 2021/2022)	0.045	0.057	0.007	0.055	0.078	0.033	0.134	0.125	-0.014
	0.162	0.180	0.719	0.182	0.183	0.500	0.293	0.276	0.689
Share of Multi-level Grades (%, 2021/2022)	0.027	0.036	0.005	0.033	0.029	-0.006	0.111	0.095	-0.022
	0.162	0.173	0.775	0.171	0.169	0.900	0.294	0.261	0.508
# of Schools	150	150	.	35	35	.	135	135	.
Joint Wald test p-value	.	.	0.004	.	.	0.107	.	.	0.256

Note: This table compares the school level baseline covariates across PSP and comparison schools for each cohort in our sample. For each group we present both the mean and the standard deviation. The difference is the difference in means controlling for regional fixed effects. Below the difference is the heteroskedastic corrected p-value. At the bottom of the table, we report the sample size of schools for each group. And add the bottom of the differences columns we report a joint Wald test p-value for the joint significance that all reported covariates are jointly different across groups.

F External Validity

F.1 Balance on Observables

Table 9: *External Validity of Schools in Study Sample*

	Non-Study	Cohort 1	Difference	Cohort 2	Difference	Cohort 3	Difference	All Cohorts	Difference
Grade 6 Final Exam Score (avg. per classroom, 2021/2022)	5.744	5.623	-0.001	5.440	-0.000	5.654	0.000	5.654	-0.001
	1.076	0.984	0.230	1.179	0.368	1.047	0.313	1.020	0.107
Grade 6 Final Exam Score (avg. per classroom, 2020/2021)	5.722	5.610	-0.001	5.372	-0.000	5.622	0.000	5.613	-0.002
	1.146	1.057	0.080	1.035	0.155	1.076	0.850	1.067	0.022
Grade 6 Final Exam Score (avg. per classroom, 2018/2019)	5.469	5.572	0.001	5.237	-0.000	5.489	0.001	5.540	0.001
	1.224	1.054	0.026	1.336	0.745	1.188	0.110	1.122	0.072
Grade 6 Final Exam Score (avg. per classroom, 2017/2018)	5.279	5.309	0.000	5.141	0.000	5.356	0.001	5.354	0.001
	1.273	1.126	0.393	1.218	0.942	1.154	0.003	1.149	0.065
Grade 6 Final Exam Score (avg. per classroom, 2016/2017)	4.516	4.631	0.001	4.714	0.000	4.653	0.001	4.671	0.002
	1.386	1.174	0.105	1.282	0.010	1.269	0.052	1.231	0.001
Grade 6 Final Exam Score (avg. per classroom, 2015/2016)	4.371	4.568	0.001	4.489	0.000	4.467	0.001	4.550	0.003
	1.407	1.225	0.001	1.234	0.038	1.274	0.076	1.257	0.000
Total Enrollment (2021/2022)	194.937	503.073	0.000	490.757	0.000	449.493	0.000	475.618	0.000
	253.353	322.179	0.000	327.316	0.000	350.363	0.000	333.559	0.000
Female Enrollment (%, 2021/2022)	0.480	0.482	0.006	0.482	0.000	0.481	0.002	0.482	0.008
	0.073	0.033	0.189	0.033	0.809	0.037	0.714	0.036	0.299
Students Repeating (%, 2021/2022)	0.089	0.075	-0.032	0.084	-0.002	0.082	-0.010	0.078	-0.055
	0.062	0.037	0.000	0.039	0.480	0.042	0.092	0.038	0.000
Student Dropout (%, 2021/2022)	0.010	0.011	-0.005	0.010	-0.013	0.011	-0.000	0.011	-0.007
	0.021	0.014	0.801	0.011	0.013	0.015	0.989	0.013	0.839
Students Qualifying for Tayssir (%, 2021/2022)	0.565	0.205	-0.045	0.230	-0.008	0.358	-0.012	0.265	-0.079
	0.290	0.298	0.000	0.310	0.000	0.328	0.000	0.320	0.000
Students Qualifying for Social Security (%, 2021/2022)	0.569	0.206	-0.044	0.230	-0.008	0.359	-0.012	0.267	-0.078
	0.293	0.300	0.000	0.310	0.000	0.329	0.000	0.322	0.000
Urban	0.147	0.660	0.043	0.629	0.008	0.415	0.010	0.568	0.071
	0.355	0.475	0.000	0.487	0.000	0.494	0.000	0.496	0.000
Number of Teachers	7.895	18.547	0.001	18.771	0.000	16.926	0.001	17.787	0.003
	8.335	10.088	0.000	9.690	0.000	11.179	0.000	10.522	0.000
School's Inspector is Trained	0.075	0.827	0.098	0.157	0.001	0.385	0.008	0.552	0.141
	0.263	0.379	0.000	0.367	0.452	0.488	0.008	0.498	0.000
School Type: Community	0.013	0.013	-0.000	0.029	-0.002	0.015	-0.003	0.014	0.001
	0.112	0.115	0.996	0.168	0.000	0.121	0.463	0.116	0.935
School Type: Central/Sector	0.196	0.283	0.002	0.286	-0.000	0.470	0.014	0.348	0.024
	0.397	0.451	0.175	0.455	0.386	0.500	0.000	0.477	0.000
Share of Multi-level classrooms (%, 2021/2022)	0.541	0.051	-0.022	0.067	-0.004	0.129	-0.011	0.098	-0.050
	0.452	0.171	0.000	0.181	0.000	0.284	0.000	0.245	0.000
Share of Multi-level Grades (%, 2021/2022)	0.517	0.031	-0.020	0.031	-0.004	0.103	-0.010	0.073	-0.045
	0.475	0.167	0.000	0.169	0.000	0.277	0.000	0.240	0.000
# of Schools	20566	300	.	70	.	270	.	640	.
Joint Wald test p-value	.	.	0.000	.	0.033	.	0.000	.	0.000

Note: This table compares the school level baseline covariates across non-study schools and the schools selected for each cohort sample. For each group we present both the mean and the standard deviation. The difference is the difference in means controlling for regional fixed effects. Below the difference is the heteroskedastic corrected p-value. At the bottom of the table, we report the sample size of schools for each group. And add the bottom of the differences columns we report a joint Wald test p-value for the joint significance that all reported covariates are jointly different across groups.

G Untestable Hypotheses

Alongside our mechanism analysis, we discuss several hypotheses that cannot be empirically tested with the available data.

Implementation intensity and teacher incentives. The implementation of the PSP may have differed between the first and second years as the program scaled from a relatively small rollout (626 schools) to a much larger national expansion (2,626 schools). We identify four dimensions along which implementation differed across these two years. While we cannot test the magnitude of these differences with the available data, access to school or classroom-level implementation data would allow us to assess whether these dimensions changed meaningfully from Year 1 to Year 2.

1. **Teacher Training Quality and Intensity.** According to implementation teams, teacher training and support may have been more 'intensive' in Year 1, when the implementation sample was smaller. This may be reflected in both the selection of trainers (inspectors) – who were among the most qualified and motivated within the inspectorate – and the duration and structure of their training. The Year 1 training cycle was longer and included a hands-on practicum with students, allowing inspectors – who are typically not in direct contact with classrooms on a daily basis – to practice and internalize the pedagogical techniques before supporting teachers. In contrast, the Year 2 training was shorter and did not include a practicum component, potentially reducing inspectors' capacity to support teachers.

2. **Frequency and Quality of School Support Visits.** According to implementation teams, inspectors conducted more school visits in Year 1 than in Year 2. As a result, the intensity and quality of teacher support were higher in the first year, providing more consistent feedback and reinforcement of program practices.

3. **Incentive Structure for Participating Schools.** The PSP included an annual financial incentive for the schools implementing the program. This financial incentive was awarded to schools in Year 1. However, this incentive applied only to Year 1, which may have affected teachers' motivation and engagement in the second year.

   On the other hand, in Year 1, a national teacher strike occurred, to which PSP teachers might have been less likely to participate in, if they were satisfied with the financial incentive their school received that year. However, initial evidence collected by the implementing teams suggested that strike participation rates were similar in both PSP and comparison schools.

4. **Availability of Instructional Materials.** One improvement in Year 2 compared to Year 1 was that textbooks and teaching materials were available from the start of the school year. In Year 1, these materials arrived later, which may have affected early implementation fidelity and limited students' access to textbooks at home during the initial months of the program. This factor would run counter to our results, as earlier availability of instructional materials would be expected to strengthen learning gains in

Year 2. While this pattern is not observed in the data, it is possible that any benefits from the improved availability of learning materials were offset by other factors in the second year.

Spillovers to comparison schools. Another possibility is that comparison schools experienced direct or indirect exposure to elements of the PSP between Years 1 and 2, for example through teacher mobility or informal diffusion of pedagogical practices. Such spillovers could, in principle, attenuate estimated treatment effects by narrowing differences between PSP and comparison schools. However, we do not have data on teacher transfers or classroom practices and therefore cannot assess the presence or magnitude of such spillovers. Moreover, given the multi-dimensional nature of the PSP – which combines structured pedagogy, targeted remediation, teacher training and specialization, and material support – it is unclear to what extent partial or indirect exposure to the program alone could explain the apparent "catch up" of the comparison schools to the PSP schools in year 2.